

DATA COMMUNICATION AND COMPUTER NETWORKING

CHAPTER ONE-INTRODUCTION TO COMPUTER NETWORKS

What is a computer network ?

A computer network is a system in which a number of independent computers are linked together to share data and peripherals , such as files and printers.

Advantages and Disadvantages of Computer Networks

The reason for the popularity of computer networks is that they offer many advantages. Information such as important files, video and audio , and email can be easily shared between users. Peripherals such as printers and modems can also be shared over the network. **For example :- the figure below shows a printer being used in a stand alone environment and in a network environment.** By connecting many computers to a print server any of them may make use of the printer directly , instead of the single computer in the stand – alone environment. Also , software such as word processors and spreadsheets can be made available to all computers on the network from a single central server.

This type of set up is known as a **client – server model**. In this model, there is a central server that provides access to various resources , such as software applications and files , to multiple client devices on the network. The client devices can use these resources remotely , without the need for installing the software or storing the files locally on their own devices , This approach can provide various benefits such as improved resource utilization , centralized management , and enhanced security.

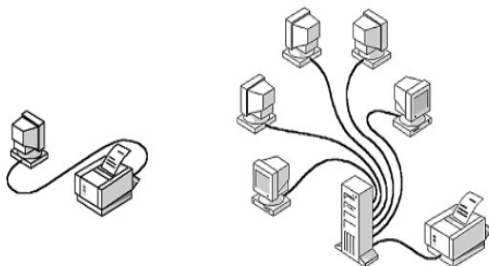


Figure 1 – A printer in a stand-alone environment (left) and in a networked environment (right)

However , with these advantages come a number of potential disadvantages. Making important and sensitive information available to every user of the network is not normally desirable. For example information about employees salaries should not be freely available for any body to look at.

Security is therefore an important concern in a network environment , secondly the danger of computer viruses entering the network is greatly increased. A virus can infect any of the computers on the network , and can quickly spread throughout the network causing significant damage.

COMPUTER NETWORK TYPES :-

Local Area Networks :-

A local area network (LAN) is a computer network that connects devices with in a limited area such as a home , office or building. It allows devices to communicate with each other and share resources such as printers , files and internet connections.

LAN can be set up in a variety of ways , including using Ethernet cables to connect devices directly to a router or switch , using wireless technologies such as wifi to connect devices to a central access point , or using power line networking to transmit data over electrical wiring.

LANs are often used in business to connect computers and other devices in an office or building , allowing employees to share resources and work collaboratively. They are also commonly used In homes to connect devices such as computers , printers , and smart devices.

LANs are typically faster and more secure than wide area networks (WANs) because they are confined to a smaller physical area and can be more easily managed and secured.

Wired LANs use a range of different transmission technologies , most of them use copper wires , but some use optical fiber. LANs are restricted in size , which means that the worst case

transmission time is bounded and known in advance.

Metropolitan Area Networks :-

A metropolitan area network (MAN) is a computer network that covers a larger geographical area than a local area network (LAN) but smaller than a wide area network (WAN). A MAN typically spans a city or a region and connects multiple LANs within that area. A MAN is typically owned by a single organization or service provider, such as a government agency or tele-communications company, and is used to connect multiple sites or buildings in a metropolitan area.

MANs can be set up using a variety of technologies including Ethernet, fiber-optic cables, microwave links, and wireless technologies such as Wifi. They are typically faster and more reliable than WANs, which cover larger geographic areas and may require leased lines or other technologies to connect remote locations.

MANs are commonly used by business and organizations that have multiple sites or branches in a metropolitan area, such as universities, hospitals, government agencies and large corporations. They allow these organizations to share resources and data across multiple locations, improving collaboration and productivity.

Wide Area Networks :-

A wide area network (WAN) is a computer network that spans a large geographical area, such as a country or even the world. WANs typically connect multiple LANs and MANs, allowing devices and users in different locations to communicate and share resources.

The main difference between LANs, MANs, and WANs is their size, scope and the technologies used to connect devices. Here are some of the key differences and similarities.

1) Size :- LANs are the smallest of the three networks, typically covering a single building or campus. MANs cover a larger geographic area such as a city or region, and WANs can span countries or even the world.

2) Scope :- LANs and MANs are typically owned and operated by a single organization or service provider, whereas WANs may be operated by multiple organizations or service providers.

3) Technologies :- LANs and WANs can use a variety of wired and wireless technologies to connect devices, including Ethernet, WiFi, and fiber-optic cables. WANs typically use specialized technologies such as leased lines, satellite links and VPNs to connect and remote locations.

4) performance :- LANs and MANs typically offer higher performance and lower latency (latency refers to the delay that occurs when data is transmitted between devices over a network) than WANs due to their smaller geographic area and more direct connections. WANs may experience greater latency and slower speeds due to the longer distances and more complex routing involved in connecting remote locations.

NETWORK CONFIGURATION :-

All networks have certain components, functions and features in common, these include :-

Servers – computers that provide shared resources for network users

Clients – computers that access shared resources provided by servers

Shared data :- files provided to clients by servers across the network

Shared peripherals :- additional hardware resources provided by servers

Even with these similarities, networks are divided into two broad categories

- peer to peer networks
- server based networks

Factors that must be evaluated to configure a network as peer to peer or server based

The type of network you choose to implement will depend on factors such as the

- Size of the organization
- Level of security required

- Types of business
- Level of administrative support available
- Amount of network traffic
- Needs of the network users
- Network budget

Peer – to – Peer Networks

In a peer to peer network , there is no dedicated servers , and there is no hierarchy among the computers. All the computers are equal and therefore are known as peers. Each computer functions as both a client and a server , and there is no administrator responsible for the entire network. The user at each computer determines what data on that computer is shared on the network.

Peer to peer networks are relatively simple , because each computer functions as a client and as a server , there is no need for a powerful central server or for other components required for a high capacity network. Peer to peer networks can be less expensive than server-based networks.

Peer to peer networks are good choices for environments where :-

- There are 10 users or fewer
- Users share resources , such as printers but no specialized servers exist
- Security is not an issue

Server Based Networks :-

In an environment with more than 10 users , a peer-to-peer network – with computers acting as both servers and clients will probably not be adequate. Therefore , most networks have a dedicated servers. A dedicated server is one that functions only as a server and is not used as a client.

As networks increase in size (as the number of connected computers and the physical distance and traffic between them , grows) more than one server is usually needed. Spreading the networking tasks among several servers ensures that each task will be performed as efficiently as possible.

A network may have separate servers for file storage , printing , email and for storing and running application software.

- 1) File and Print servers
- 2) Application servers
- 3) Mail servers
- 4) Fax servers
- 5) Communication servers
- 6) Directory services

Although it is more complex to install , configure ,and manage , a server - based network has many advantages over a simple peer to peer network.

Number of users :- It supports large number of users

Server-based data sharing can be centrally administered and controlled :-, because these shared resources are centrally located , they are easier to find and support than resources on individual computers.

Security :- is often the primary reason for choosing a server – based approach to networking. In server-based environment , one administrator who sets the policy and applies it to every user on the network can manage security.

Backups can be scheduled several times a day or once a week depending on the importance and value of the data. Server back ups can be scheduled to occur automatically , according to a predetermined schedule , even if the servers are located on different parts of the network.

A server – based network can support thousands of users. This type of network would be impossible to manage as a peer to peer network.

impossible to manage as a peer-to-peer network.

Comparison of Network Types

Consideration	Peer-to-Peer Network	Server-Based Network
Size	Good for 10 or fewer computers	Limited only by server and network hardware
Security	Security established by the user of each computer	Extensive and consistent resource and user security
Administration	Individual users responsible for their own administration; no full-time administrator necessary	Centrally located for network control; requires at least one knowledgeable administrator

ADVANTAGE AND DISADVANTAGE OF PEER TO PEER NETWORK

ADVANTAGE :-

- ◆ Allows simple sharing of data and hardware
- ◆ Not expensive
- ◆ Easy to set up

DISADVANTAGE :-

- ◆ Poor security of data and resources
- ◆ Not suitable for large number of PCs

ADVANTAGES OF CLIENT / SERVER NETWORK

Sharing Resources :- because shared resources are centrally located , they are easier to find , administer and control

Security :- Security is often the primary reason for choosing this network model , it has got good security data and other resources

Backup :- Backups can be scheduled several times a day or a week depending on the importance and value of the data

Number of users :- It supports large number of users

Computer Network Topologies

The term topology or more specifically , **network topology refers to the arrangement or physical layout of computers , cables and other components on the network.** A network's topology affects its capabilities. The choice of one topology over another will have an impact on the :-

- Type of equipment that the network needs
 - Capabilities of the network
 - Growth of the network
 - Way of the network managed

Different topologies require different communication methods and these methods have a great influence on the network.

There are four basic types of computer topology : bus , star , ring , mesh , hybrid.

1) Bus Topology :-

Bus Topology is a type of network topology in computer networking where all the nodes in a network are connected to a common linear transmission medium , which is typically a coaxial cable or twisted pair cable. In a bus topology , the cable acts as back bone for the network , and all the nodes are attached to the cable via a drop or a T-connector.

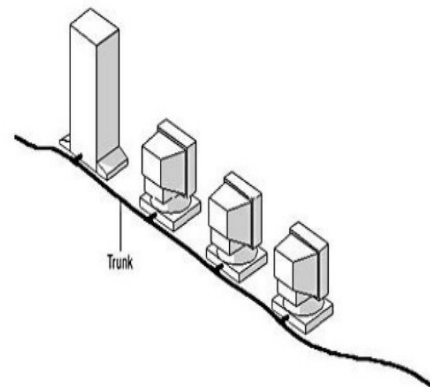


Figure 1- The bus topology

The bus topology is known for its simplicity and ease of installation , as it does not require any central point of control or complex network management. It is also cost – effective , as it requires less cabling than other topologies , making it suitable for small networks.

However , the bus topology has some drawbacks. One of the main disadvantage is that the entire network is reliant on a single cable , which makes the network vulnerable to a single point of failure. If the cable is damaged or breaks , the entire network will be down. Additionally , the performance of the network may be affected if there are too many nodes connected to the network , as the band width of the cable is shared among all of the nodes. Bus topology is a simple and cost-effective way to connect nodes in a network , but it may be the most reliable or scalable option for a larger networks.

2) Star Topology

In the star topology, cable segments from each computer are connected to a centralized component called a hub. Figure 2 shows four computers and a hub connected in a star topology. Signals are transmitted from sending computer through the hub to all computers on the network.

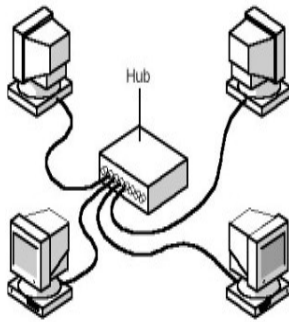


Figure 2 – The star topology

Because each computer is connected to a central point, this topology requires a great deal of cable in a large network installation. Also, if the central point fails, the entire network goes down. If one computer or the cable that connects it to the hub fails on a star network only the failed computer will not be able to send or receive network data. The rest of the network continues to function normally.

Advantage of star topologies :-

- Like a bus topology, networks are easy to install and maintain
- If there is any interruption between any of the clients and the hub, the other clients will normally work
- Since there is a dedicated controller, the hub, collision is much reduced
- Nodes can be added to and removed from the network with little or no disruption to the network.

Disadvantage of star topologies

- We need to use more cable
- If the hub fails, then the whole network will fail, so the entire network will depend on hub.

Ring Topology

The ring topology connects computers on a single circle of cable. Unlike the bus topology, there are no terminated ends. The signals travel around the loop in one direction and pass through each computer, which can act as a repeater to boost the signal and send it on the next computer. Figure 3 shows a typical ring topology with one server and four work stations. The failure of one computer can have an impact on the entire network.

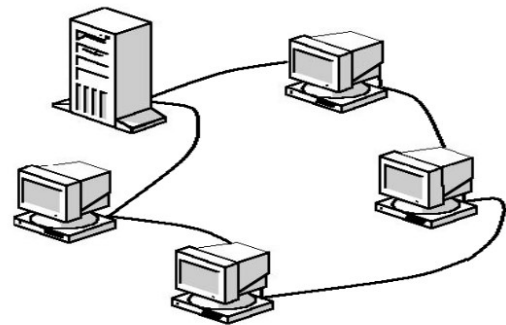


Figure 3 – The ring topology

Each computer acts as a repeater, regenerating the signal and sending it on to the next computer, thereby preserving signal strength. One disadvantage is that if there is an interruption at some point, the whole network will go down.

Mesh Topology

A mesh topology network offers superior redundancy and reliability. In a mesh topology, each computer is connected to every other computer by separate cabling. This configuration provides redundant paths throughout the network so that if one cable fails, another will take over the traffic. While ease of troubleshooting and increased reliability is definite benefits, these networks are expensive to install because they use a lot of cabling. Often a mesh topology will be used in conjunction with other topologies to form a hybrid topology.

An advantage of this topology is its backup capabilities by providing multiple paths through the network, because redundant paths require more cable than needed in other topologies, a mesh topology can be expensive.

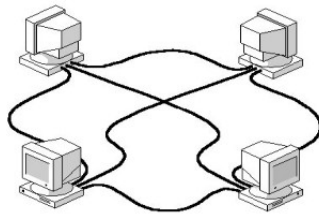


Figure 4 – The mesh topology

Hybrid Topology :- Many working topologies are hybrid combinations of the bus , star , ring and mesh topologies. Two of the common are described below.

Start bus topology :- The star bus is a combination of the bus and star topologies , several **start topology** networks are linked together with **linear bus trunks**.

DATA REPRESENTATION TECHNIQUES

Data representation refers to the methods used to represent different types of information in a computer system. Computers store and process data in various formats, including numeric data, text, executable files, images, audio, video, and more. However, these different types of data appear distinct to us as humans.

Here are some common methods of data representation in data communication:

1. **Decimal Numbers:** Decimal numbers are the numbers we are most familiar with in our everyday lives. They use a base-10 system, meaning they consist of ten digits (0-9). Decimal numbers represent quantities in terms of units, tens, hundreds, and so on.
2. **Binary Numbers:** Binary numbers use a base-2 system and consist of only two digits, 0 and 1. Computers internally represent and process data in binary format. Each digit in a binary number is called a bit, and a sequence of bits forms bytes, which are the basic units of storage in computers.
3. **Hexadecimal Numbers:** Hexadecimal numbers use a base-16 system, combining

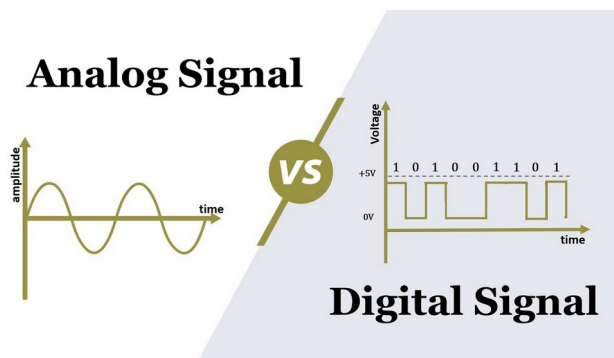
digits from 0 to 9 and letters from A to F. They are commonly used in computer programming and data representation, as they provide a more compact and convenient way to represent binary data. Hexadecimal numbers are often used to represent memory addresses, color codes, and machine instructions.

4. **Text:** Text data represents characters and symbols using a standardized encoding system, such as ASCII (American Standard Code for Information Interchange) or Unicode. Each character is assigned a unique numeric value, allowing computers to store, process, and display text-based information. Text data can include letters, numbers, punctuation marks, and special characters. For example, the uppercase letter 'A' is assigned the code 65, the lowercase letter 'a' is assigned the code 97, and so on.
5. **Graphics:** Graphics data represents visual images, such as drawings, photographs, icons, and animations. Graphics are typically stored in formats like JPEG, PNG, GIF, or BMP. These formats use various techniques to encode and compress visual information, allowing it to be stored and displayed on computer screens.

Each method of data representation has its own advantages and use cases. The appropriate representation depends on the type of data being handled and the specific requirements of the application or system. Effective data representation is essential for accurate communication and efficient processing of information in computer systems.

Data Transmission formats

Data transmission refers to the process of sending digital or analog data from one device to another over a communication medium. **It allows devices to communicate with each other by transferring data in the form of signals.** Data transmission can occur between devices in various ways, such as point-to-point (between two devices), point-to-multipoint (from one device to multiple devices), or multipoint-to-multipoint (between multiple devices).



Digital data transmission involves sending and receiving digital data, which is represented by discrete signals or digital bit streams. The data originates from a source device and is transmitted over a communication medium, such as copper wires, wireless carriers, or optical fiber, to reach the destination device. The digital data is composed of binary digits (bits), which can be either 0 or 1. These bits are combined to represent different types of information, such as text, images, audio, or video.

Analog transmission, on the other hand, involves conveying voice, data, images, signals, or video information using continuous signals that vary in amplitude, phase, or another property. An analog signal is characterized by its continuous variation along amplitude and frequency. For example, in telephony, when you speak into a handset, the changes in air pressure are converted into electrical signals that represent the voice pattern. These signals are analog in nature because they continuously vary.

Digital transmission is different from analog transmission in several ways. Digital signals are simpler and consist of a series of discrete pulses that represent 0s and 1s (bits). Each computer uses a coding scheme to define how different combinations of 0s and 1s represent characters in a character set, such as letters, numbers, punctuation marks, and control functions. Digital transmission allows for more reliable and accurate data transfer compared to analog transmission.

Data transmission is a crucial process that enables devices to communicate and share information, whether in digital or analog form, using various methods and transmission media.

TRANSMISSION IMPAIRMENTS

When we transmit signals over a communication link, there can be some problems that affect the quality of the signal. These problems are known as **transmission impairments** and can occur in both analog and digital signals. There are three main types of transmission impairments: attenuation, delay distortion, and noise.

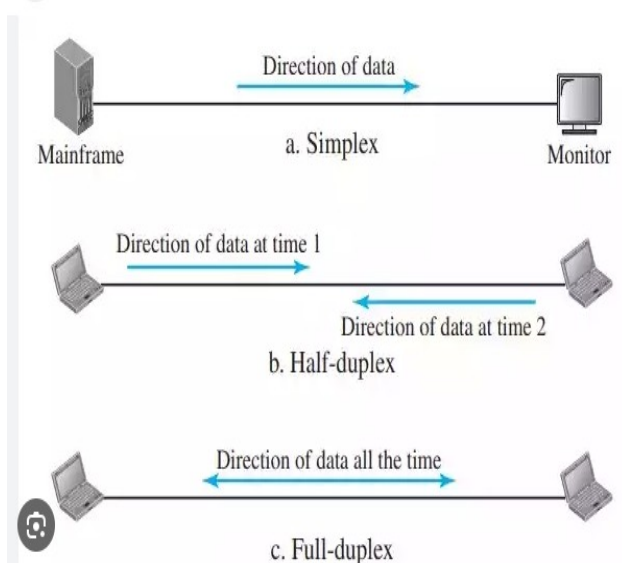
- 1. Attenuation:** Attenuation happens when the strength of the signal decreases as it travels over a distance. This can make the signal weaker and harder to understand.
 - We need the received signal to be strong enough for the receiver to understand it. We can use amplifiers or repeaters to boost the signal strength.
- 2. Delay Distortion:** Delay distortion occurs because different frequencies in a signal can travel at slightly different speeds through a transmission medium. This can cause the signal to become distorted or arrive at different times. To fix this, we can use equalization techniques to smooth out the distortion.
- 3. Noise:** Noise refers to unwanted signals that interfere with the transmission and reception of the intended signal. There are different types of noise:

- **Thermal noise:** This noise is caused by the random motion of electrons due to temperature. It cannot be eliminated completely and is directly related to the temperature and bandwidth of the system.
- **Crosstalk:** Crosstalk occurs when signals from one communication channel leak into another nearby channel. This can happen in twisted pair cables or microwave antennas and can cause interference or overlap of signals.

Impairments can affect the quality and reliability of data transmission. To mitigate their effects, various techniques and technologies are used, such as amplifiers, equalizers, and error-correcting codes. By understanding these impairments and implementing appropriate solutions, we can improve the performance of communication systems.

DATA TRANSMISSION MODES

Data transmission mode refers to the direction in which information flows between two communication devices. It determines how data is exchanged and synchronized in a computer network.



There are three main data transmission modes based on the direction of information exchange:

1. **Simplex:** In simplex mode, data can only flow in one direction, which means the communication is unidirectional. The sender can only send data, while the receiver can only receive it. Two-way communication is not possible in this mode. Simplex mode is commonly used in situations where there is no need for a corresponding reply. Examples include radio and TV transmissions, keyboards, and mice.

Keyboards and mice generally use simplex communication, which is a type of data transmission mode where data only travels in one direction - from the device (keyboard or mouse) to the computer.

In a typical computer setup, the mouse and keyboard don't need to receive data from the computer under normal operation - they only need to send signals to it. For example, when you press a key on your keyboard, it sends a signal to your computer indicating which key was pressed. **The computer then interprets this signal and carries out the appropriate action (well it will do some operation but it does not reply back to the keyboard).** The same thing happens when you move your mouse or click one of its buttons.

However, in some more advanced or specialized situations, a keyboard or mouse might also receive data from the computer, such as for configuration purposes or to control lighting on RGB keyboards and mice. In these cases, the communication would be duplex, not simplex. Still, for the majority of usage, keyboards and mice operate using simplex communication.

2. **Half-Duplex:** In half-duplex mode, data can flow in both directions, but only in one direction at a time. Also known as semi-duplex, each station can both transmit and receive data, but not simultaneously. When one device is sending data, the other can only receive,

and vice versa. The entire channel capacity is utilized for each direction of communication. This mode is suitable when communication in both directions is not required simultaneously. Examples include police radios and internet browsers.

3. **Full-Duplex:** Full-duplex mode enables data to flow in both directions simultaneously, allowing for bi-directional communication. Both stations can transmit and receive data at the same time. Full-duplex mode provides double the bandwidth compared to half-duplex mode. The channel capacity is divided between the two directions of communication. This mode is used when simultaneous communication in both directions is required. An example is a telephone network, where two people can talk and listen to each other simultaneously.

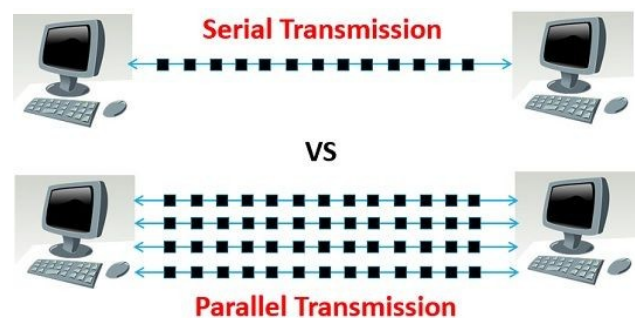
The choice of data transmission mode depends on the specific communication requirements of a system. Simplex mode is used for one-way communication, while half-duplex mode suits situations where bidirectional communication is not needed simultaneously. Full-duplex mode is ideal for scenarios where simultaneous two-way communication is required.

In synchronous transmission, data is sent continuously without any breaks or pauses. It's like a synchronized dance where both the sender and receiver follow the same rhythm or timing. They are always in sync with each other, so they know exactly when the data is being transmitted.

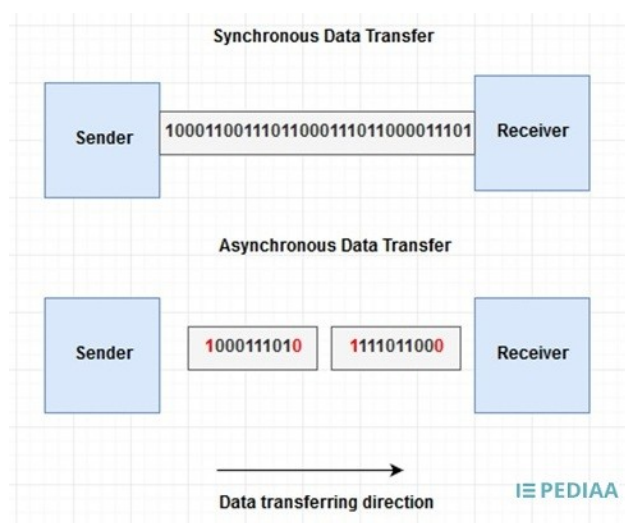
In asynchronous transmission, data is sent in smaller chunks with specific start and stop points. It's like sending messages with clear beginnings and endings. The sender adds a start bit at the beginning and a stop bit at the end of each chunk of data to indicate when the transmission starts and ends.

To simplify further, synchronous transmission is like a smooth and continuous flow of data, while asynchronous transmission is like sending separate messages with clear markers to indicate the start and end of each message.

SERIAL AND PARALLEL TRANSMISSION



Synchronization and Asynchronous Transmission



Serial transmission: In this mode, data is sent one bit at a time over a single transmission line. It's like sending a series of letters, one after the other, through a narrow tunnel. The sender and receiver need to be synchronized to ensure the data is received correctly. This mode is used for long-distance communication or when the amount of data being sent is small.

Parallel transmission: In this mode, multiple bits are sent at the same time using multiple transmission lines. It's like sending a group of letters together through separate tunnels. Parallel transmission allows for faster data transfer because more bits can be transmitted simultaneously. However, it requires more transmission lines and can be affected by

interference if the lines are too close together. It is used for short-distance communication or when a large amount of data needs to be sent quickly.

When selecting a data transmission mode, several factors need to be considered:

- **Transmission rate:** How quickly data can be sent and received.
- **Distance:** The maximum distance over which data can be reliably transmitted.
- **Cost and ease of installation:** The expense and difficulty involved in setting up the transmission system.
- **Resistance to environmental conditions:** The ability of the transmission system to withstand and perform well in different environments.

By considering these factors, the most appropriate transmission mode can be chosen for a specific network communication requirement.

ELEMENTS OF A DATA COMMUNICATION

Message: The message is the information or data that needs to be communicated. It can be in various forms such as text, audio, video, or any other type of data.

Sender: The sender is the device or entity that initiates the communication by transmitting the message. It can be a computer, mobile phone, workstation, or any other device capable of sending data.

Receiver: The receiver is the device or entity that receives the message sent by the sender. It is the destination where the message is intended to be received. Similar to the sender, the receiver can be a computer, mobile phone, workstation, or any other device capable of receiving data.

Transmission Medium: The transmission medium is the physical path or channel through which the data or message travels from the sender to the receiver. It can be either guided (using wired connections like cables) or unguided (using wireless connections like radio waves or

microwaves). Examples of transmission media include twisted pair cables, fiber optic cables, and wireless communication channels.

Set of Rules (Protocol): In order for communication to take place successfully, a set of rules and procedures called a protocol needs to be established. Protocols define how the sender and receiver should communicate, including the format of the message, error handling, data flow control, and other aspects of the communication process. Without a protocol, devices may not be able to understand each other and communication would not be possible.

The elements mentioned above work together to enable effective data communication between devices. For example, when sending an email, the sender acts as the source, the message is the content of the email, the receiver is the intended recipient, the internet connection serves as the transmission medium, and protocols like SMTP (Simple Mail Transfer Protocol) govern the rules for sending and receiving emails.

In computer networks, protocols are essential to ensure reliable and standardized communication between devices, allowing for seamless data exchange and interoperability.

TYPES OF STANDARDS

De Facto Standard: These are standards that have not been officially approved by any organization but are widely adopted due to their widespread use or industry acceptance. They are often established by manufacturers or companies as their own rules or conventions.

De Jure Standard: These are standards that have been officially approved by recognized bodies or organizations such as ANSI (American National Standards Institute), ISO (International Organization for Standardization), IEEE (Institute of Electrical and Electronics Engineers), etc. They are important to follow when required or needed, especially in data communication protocols.

Circuit Switching and Packet Switching



**Circuit
Switching**

Vs



**Packet
Switching**

Circuit switching is a communication method where a dedicated communication path or circuit is established between two devices before data transmission begins.

Advantages:

Guaranteed bandwidth: The dedicated path ensures a specific amount of bandwidth for the duration of the communication.

Low latency: Since the path is predetermined, there is no need to establish a connection for each data packet, resulting in low latency.

Predictable performance: The reserved bandwidth ensures predictable performance without resource competition.

Suitable for real-time communication: Circuit switching is suitable for real-time applications like voice and video due to low latency and guaranteed bandwidth.

Disadvantages:

Inefficient use of bandwidth: The dedicated circuit reserves bandwidth even when there is no data transmission, resulting in inefficient utilization.

Limited scalability: The number of circuits that can be established is finite, which limits the number of simultaneous calls that can be made.

High cost: Circuit switching requires dedicated resources, making it more expensive compared to other methods.

Packet Switching:

Packet switching is a communication method where data is divided into smaller units called packets and transmitted over the network.

Advantages:

Efficient use of bandwidth: Bandwidth is shared among multiple users, and resources are allocated only when data needs to be transmitted, resulting in efficient utilization.

Flexible: Packet switching can handle a wide range of data rates and packet sizes.

Scalable: It can handle large amounts of traffic on a network, making it highly scalable.

Lower cost: Resources are shared among multiple users, reducing overall costs.

Disadvantages:

Higher latency: Due to the need for packet routing through multiple nodes, packet switching has higher latency compared to circuit switching.

Packet loss: Congestion or transmission errors can result in packet loss during transmission.

Unsuitable for real-time communication: Packet switching may not be suitable for real-time applications like voice and video due to potential latency and packet loss.

Similarities:

- ➔ Both circuit switching and packet switching involve the transmission of data over a network.
- ➔ Both methods use the physical layer of the OSI model for data transmission.
- ➔ Both methods can be used to transmit voice, video, and data.

- ➔ Both methods can be used within the same network infrastructure.
- ➔ Both methods can be used for both wired and wireless networks.

These concepts are important in understanding the different modes of data communication and the trade-offs associated with each method.

Computer Network Components

Computer Network Components Repeaters

A computer network repeater is a device used to boost or regenerate digital signals on a network. It receives a signal on one port and amplifies it before retransmitting it on one or more output ports. The primary function of a repeater is to extend the range of a network by amplifying the signal so that it can travel further along the network's physical medium, such as a copper wire or fiber optic cable.

It is important to note that repeaters do not improve the quality of the signal, and they can not correct errors or noise that may present in the original signal, instead they simply regenerate the signal and forward it along the network.

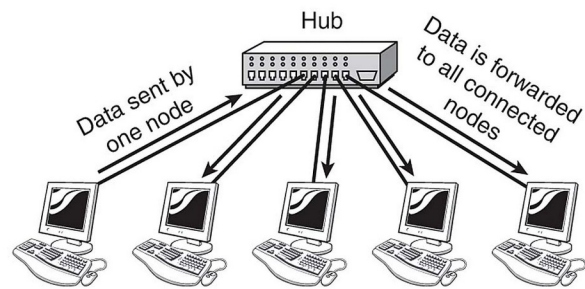
Hubs

A hub is a networking device that serves as a central connection point for multiple devices on a network. It receives data packets from one device and broadcasts them to all the devices connected to it, regardless of whether the data is intended for a specific device or not.

Hubs work by receiving data packets on one port and then broadcasting them to all other ports on the hub, regardless of whether the data is intended for specific device or not. This is called a "broadcast" or "collision" domain. As a result, all devices connected to the hub receive the same data packets, and each device must determine whether the packet is intended for it or not.

There are two types of hubs: **passive** and **active**. A passive hub simply acts as a connector for devices and does not amplify or regenerate the

signals. An active hub, on the other hand, includes electronics that amplify and regenerate the signals to boost the strength and quality of the transmission.



Hubs have largely been replaced by switches, which are more efficient and intelligent in handling network traffic. Switches are able to send data only to the device that it is intended for, which reduces network congestion and improves network performance.

Because hubs broadcast all data packets to all connected devices, they can cause network congestion and reduce network performance, particularly in larger networks.

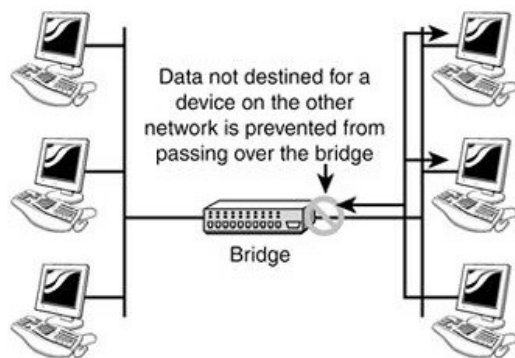
Bridges

A bridge is a networking device that connects two separate network segments to create a single, larger network. Bridges work by examining the source and destination MAC addresses of data packets and forwarding them only to the segment where the destination MAC address is located. This helps to reduce network congestion and improve network performance by limiting the amount of unnecessary traffic that is sent across the network.

Bridges are particularly useful in large networks that are divided into multiple segments as they can help to improve the efficiency of the network by reducing the amount of broadcast traffic that is sent across the entire network.

When a device on one segment of a network wants to send data to a device on another segment of the same network, it sends a data packet containing the source and destination MAC addresses involved. The bridge examines the source and destination MAC addresses in the data packet and compares them to its internal table of

MAC addresses and corresponding segments. This table is called a forwarding table or a MAC table.



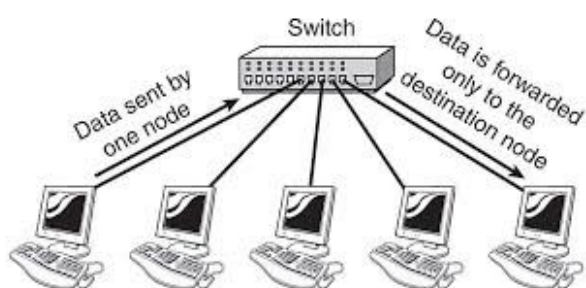
If the destination MAC addresses is not found in the forwarding table , the bridge assumes that the destination device is on a different segment of the network and forwards the data packet to all segments except the one it was received on.

When the destination device responds to the original data packet , the bridge updates its forwarding table with the MAC address and corresponding segment of the device. From that point on , any data packets destined for that device are forwarded only to the segment where the device is located , reducing the amount of unnecessary traffic sent across the network.

By limiting the amount of unnecessary traffic , bridges help to reduce network congestion and improve network performance. This is particularly useful in large networks with many devices , where broadcast traffic can quickly overwhelm the network and cause performance issues.

Switches

A switch is a networking device that connects multiple devices on a network and enables them to communicate with each other. Unlike hubs , which broadcast data packets to all devices on a network , switches are able to send data only to the device that is intended for. This helps to reduce network congestion and improve network performance by limiting the amount of

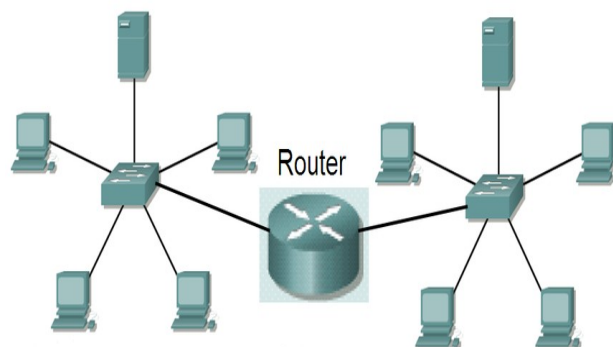


unnecessary traffic that is sent across the network.

Switches work by examining the source and destination MAC address of data packets and forwarding them only to the port where the destination MAC address is located. This process is called switching and it helps to ensure that data packets are sent only to the device that needs them.

Switches are particularly useful in large networks where there are many devices that need to communicate with each other. They can help to improve the efficiency of the network by reducing the amount of broadcast traffic that is sent across the entire network.

Routers



A router is a networking device that connects multiple networks together and enables devices on those networks to communicate with each other.

Routers work by examining the destination IP address of data packets and forwarding them to the correct network. They maintain a routing table , which contains information about the different networks that the router is connected to and the best path for sending data to each network.

Routers are particularly useful in large networks where there are multiple subnets or networks that need to communicate with each other. They can help to improve the efficiency of the network by directing traffic along the shortest and fastest path to its destination.

What makes routers differ from the switches.

Switches are networking devices that connect multiple devices on a network and enable them to communicate with each other. Switches work by examining the source and the destination MAC addresses of the data packets and forwarding them only to the port where the destination MAC address is located. This process is called switching and it helps to ensure that data packets are sent only to the device that needs them.

Routers on the other hand , connect multiple networks together and enable devices on those networks to communicate with each other. Routers work by examining the destination IP address of data packets and forwarding them to the correct network. They maintain a routing table , which contains information about the different networks that the router is connected to an the best path for sending data to each network.

The main difference between routers and switches is their functionality , switches are designed to connect devices on the same network and direct data packets to the appropriate device , while routers are designed to connect multiple networks together and direct data packers between them. As a result routers are more intelligent and capable than switches and they can make decisions about how to forward data packets based on their destination address and network topology.

Switches are designed to connect devices on the same network or subnet, such as devices on a local area network (LAN). They are able to direct data packets to the appropriate device by looking up the destination MAC address in their forwarding table, which contains a list of MAC addresses and the ports they are connected to.

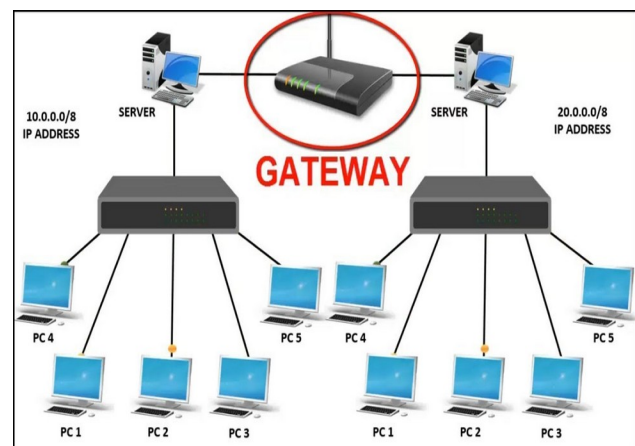
When a data packet is received on a switch port, the switch checks the forwarding table to determine which port the packet should be sent to in order to reach the destination device. This helps to reduce network congestion and improve network performance by directing traffic only to the devices that need it.

Routers, on the other hand, are more complex networking devices that operate at the network layer (Layer 3) of the OSI model. They connect

multiple networks together and enable devices on those networks to communicate with each other.

Routers work by examining the destination IP address of data packets and forwarding them to the correct network. They maintain a routing table, which contains information about the different networks that the router is connected to and the best path for sending data to each network. Routers can perform various functions such as filtering, NAT, and VPN capabilities to secure and manage network traffic.

Gateways



A gateway Is a networking device that connects two or more different networks that may use different protocols , topologies , or architectures. Its main purpose is to facilitate communication and data exchange between these networks.

In order for data to be transmitted between networks , it must be in a format that can be understood by both networks. A gateway is responsible for converting data from one network format to another. For example , if one network uses the Internet Protocol (IP) and another uses X.25 protocol , a gateway can translate data between the two protocols so that data can be transmitted between the two networks.

Gateways are often used to connect different types of networks , such as LAN and WAN or a wireless and a wired network. They can also be used connect different types of devices such as a computer and a printer or a phone and a computer.

Gateways are critical for enabling communication between different networks or devices that use different protocols , topologies , or architectures. They provide a bridge between these networks , allowing data to be transmitted and received across the different networks.

CHAPTER FOUR :- NETWORK MODELS

A network model is a conceptual framework that describes how data flows within a network and how network devices interact with each other. It provides a structured approach to designing, building and troubleshooting computer networks.

There are several network models that have been developed over time, including the open systems interconnection (OSI) model and the Transmission Control Protocol / Internet Protocol (TCP/IP) model. These models divide the networking process into layers, with each responsible for a specific set of tasks or functions.

The OSI model, for example, has seven layers, ranging from the physical layer (Layer 1) to the application layer (Layer 7). Each layer performs specific functions, such as data transmission (Layer 1), error correction (Layer 2), routing (Layer 3), and application services (Layer 7). The TCP/IP model, on the other hand, has four layers, the network interface layer, the internet layer, the transport layer, and the application layer.

The purpose of a network model is to provide a standard framework for communication between network devices, regardless of the manufacturer or type of device. This allows devices from different vendors to interoperate with each other, and amplifies the process of designing, building and maintaining networks.

Network models are used extensively in computer networking, and are a key concept for network engineers and administrators. They provide a common language and framework for discussing networking concepts and issues and are essential for troubleshooting network problems.

TYPES OF NETWORK MODELS IN DATA COMMUNICATION

The open systems interconnection (OSI) model is a conceptual framework that describes how data moves through a network. It is a seven-layer model that breaks down the network communication process into smaller and more manageable parts.

The OSI model provides a framework for understanding how network devices and applications communicate with each other. Each

layer has a specific function that contributes to the overall transmission of data between devices on a network. By breaking down the network communication process into smaller parts, the OSI model simplifies the design, implementation and troubleshooting of computer networks.

OSI is a standard description or “reference model” for how messages should be transmitted between any two points in a telecommunication network. Its purpose is to guide product implementers so that their products will consistently work with other products.

The OSI model is designed to prompt interoperability by creating a guideline for network transmission between computers that have different vendors, SW, operating systems and protocols.

The OSI reference model is the best-known and most widely used guide for visualizing networking environments. Manufacturers adhere to the OSI reference model when they designed the network products. It provides a description of how network hardware and software work together in a layered fashion to make communication possible.

A reference model is a conceptual blueprint of how communications should take place. It addresses all the processes required for effective communication and divides these processes into logical groupings called layers. The OSI model is the primary architectural model for networks. It describes how data and network information are communicated from application on one computer, through the network media, to an application on another computer. The OSI reference model breaks this approach into layers.

Major benefits of OSI reference model :-

- Reduce complexity
- Standardizes interfaces
- Facilitate-modular/modification engineering
- Ensure Interoperability Technologies
- Accelerate Evolutions
- Allow various types of network hardware and software to communicate

The OSI reference model defines how each layer communicates and works with the layers immediately above and below it.

Each of the seven layers may differ in terms of :-

Purpose / Function :- Each of these layers performs a specific function that plays an important part in the end to end communication between two devices.

Network Device :- Hubs , Switches , Bridges , Routers , NICs

Protocols :- TCP , IP , FTP , UDP , SMTP

Data handling and encryption process :- Name of the data as data , segment , Datagram , packet , frame , Bit.

The seven layers of OSI are :-

- ✓ 1) Application Layer
- ✓ 2) Presentation Layer
- ✓ 3) Session Layer
- ✓ 4) Transport Layer
- ✓ 5) Network Layer
- ✓ 6) Data Link Layer
- ✓ 7) Physical Layer

Layer 1 :- Physical Layer

The physical layer is the first layer of the OSI model and is responsible for transmitting raw data bits over a physical medium , such as copper or fiber optic cables. It defines the physical characteristics of the network , such as voltage levels , cable specifications , and data transmission rates.

The physical layer is concerned with the mechanical , electrical and physical aspects of the data transmission , and its main objective is to provide a reliable and efficient means of transmitting raw data between devices on a network.

The Physical Layer defines the physical characteristics of the network :-

- Physical characteristics can include the cable and connector type , the format for pin outs for cables and so on.

- Also defines how the data actually travels across the network
- Define the voltage used on the cable and the frequency at which the signals that carry the data are transitioned from one state to another.
- Defines the physical topology of the network.

Layer 2 :- Data Link Layer

The data link layer is the second layer of the OSI model and is responsible for error-free data transmission over a physical layer such as between network card and a switch. Its main objective is to provide a reliable and error free connection between two devices on a network.

The data link layer divides data into frames and adds error detection and correction codes to ensure that data is transmitted without errors. It also manages data flow control between devices to prevent data loss and congestion on the network.

The data link layer is responsible for tasks such as addressing , framing , error detection and correction , and flow control. It uses physical addresses , such as Media Access Control (MAC) addresses , to identify devices on a network , and is responsible for directing data frames to their intended recipients.

The data link layer is critical for ensuring that data is transmitted reliably and efficiently between devices on a network. It works closely with the physical layer to establish and maintain a reliable physical connection between devices , and is an essential component of any computer network.

The data link layer uses physical addresses such as Media Access Control (MAC) addresses , to identify devices on a network. A MAC address is a unique identifier card (NIC) on a device. These addresses are typically assigned by the manufacturer and can not be changed.

When data is transmitted between devices on a network , it is divided into frames at the data

link layer. Each frame includes a source MAC address and a destination MAC address, which are used to identify the devices that are sending and receiving the data. The data link layer is responsible for directing data frames to their intended recipients based on their destination MAC addresses.

The data link layer uses a technique called MAC address filtering to ensure that data is only sent to the devices that need it. When a device receives a data frame, it checks the destination MAC address to determine if it matches its own MAC address. If the destination MAC address matches the MAC address of the device, the data frame is accepted and processed. If the destination MAC address does not match, the data frame is discarded and not processed.

The use of MAC addresses at the data link layer is critical for ensuring that data is transmitted efficiently and securely between devices on a network. By using MAC addresses to identify devices and direct data frames to their intended recipients, the data link layer helps to prevent unnecessary network congestion, the data link layer helps to prevent unnecessary network congestion and improves overall network performance.

Layer 3 :- Network Layer

The network layer is a third layer of the OSI model and is responsible for routing data between different networks such as internet. Its main objective is to provide end-to-end data transport services and ensure that data is delivered to the correct destination device.

The network layer adds routing information to data packets and selects the best path for data transmission based on factors such as network congestion, distance, and cost. It also performs functions such as fragmentation and reassembly of data packets and provides logical addressing through the use of internet protocol (IP) addresses.

The network layer is responsible for tasks such as addressing, routing, fragmentation, reassembly and quality of service (QoS) management. It uses logical addresses such as IP addresses, to identify devices on a network, and is responsible

for directing data packets to their intended recipients.

The network layer uses a variety of routing protocols, such as Border Gateway and Open shortest path first, to select the best path for data transmission based on factors such as network congestion, distance and cost. It also performs fragmentation and reassembly of data packets when the data is too large to be transmitted in a single packet.

In addition, the network layer is responsible for QoS management, which involves managing network traffic to ensure that critical data is transmitted with priority over non-critical data. This helps to ensure that important data such as voice and video traffic, is delivered in a timely and reliable manner.

Layer 4 :- Transport Layer

The transport layer is the fourth layer of the OSI model and is responsible for providing end-to-end data transport services, such as reliable data delivery and flow control. Its main objective is to ensure that data is transmitted without errors and in the correct order between the source and destination devices.

The transport layer establishes a logical connection between the source and destination devices and provides a reliable end-to-end communication channel. It is responsible for tasks such as error detection and correction, flow control, and congestion control.

The transport layer uses transport protocols, such as Transmission Control Protocol (TCP), to provide end-to-end data transport services. TCP is a reliable transport protocol that provides error detection and correction, flow control and congestion control to ensure that data is transmitted without errors and in the correct order.

The transport layer is also responsible for managing data flow control between devices. It ensures that data is transmitted at an appropriate rate, prevents data loss and congestion on the network. This helps to ensure that data is transmitted reliably and efficiently between devices on a network.

Layer 5 :- Session Layer

The session layer is the fifth layer of the OSI model and is responsible for **establishing , maintaining and ending communication sessions between applications running on different devices**. Its main objective is to manage the flow of data between applications and provide services such as check pointing and recovery.

The session layer also provides services such as check pointing and recovery. Check pointing involves saving the state of communication session at regular intervals , so that if the session is interrupted or terminated unexpectedly , it can be restarted from the last saved checkpoint. Recovery involves resuming a communication session from the last saved check point after an interruption or termination.

Check pointing involves saving the state of communication session at regular intervals , usually at predetermined points called check points. The session layer saves important information about the session , such as the current state of the communication , the data transmitted , and the point at which the last checkpoint was taken. If the session is interrupted or terminated unexpectedly , the session layer can use the last saved check point to restart the session from the point at which the interruption occurred.

Recovery involves resuming a communication session from the last saved check point after an interruption or termination. When a communication session is interrupted or terminated , the session layer uses the last saved check point to resume the session from the point at which interruption occurred. This allows the communication session to be resumed without losing any important data or information.

Check pointing and recovery are important services for applications that require reliable and continuous communication such as file transfer or video conferencing applications. By saving the state of a communication session at regular intervals and resuming the session from the last saved checkpoint , these services ensure that communication sessions can be resumed quickly and efficiently in the event of an interruption or termination.

Check pointing and recovery are critical services provided by the session layer to ensure reliable communication between applications running on different devices. They work together to ensure that communications sessions that can be resumed quickly and efficiently , and are essential components of any computer network.

Layer 6 :- Presentation Layer

The presentation layer is the sixth layer of the OSI model and is responsible for ensuring that data is presented in a format that can be understood by the receiving device. Its main objective is to manage the syntax and semantics of the information exchanged between applications.

The presentation layer provides services such as data compression , data encryption , and data translation. Data compression involves reducing the size of the data to reduce transmission time and network bandwidth usage. Data encryption involves encoding data to protect it from unauthorized access or tampering. Data translation involves converting data between different formats such as ASCII to EBCDIC , to ensure that data can be understood by the receiving device.

The presentation layer is also responsible for data formatting and syntax control. It ensures that data is presented in a format that can be understood by the receiving device , such as by converting data into a standard format or using a common coding scheme.

The presentation layer is critical for ensuring that data is presented in a format that can be understood by the receiving device . It works closely with the application layer to manage the syntax and semantics of the information exchanged between applications , and is an essential component of any computer network.

Layer 7 :- Application Layer

The application layer is the seventh and top most layer of the OSI model and is responsible for providing communication services directly to applications running on different devices. Its main objective is to enable communication between different types of applications such as web

browsers , email clients and file transfer programs.

The application layer provides services such as email , file transfer , remote login and web browsing. Email services allow users to send and receive emails , file transfer services enable users to transfer files between devices , remote login services allow users to remotely access a device , and a web browsing services enable users to access and navigate websites on the internet.

The application layer is responsible for managing application level protocols such as HTTP , SMTP and FTP , which are used to enable communication between applications. It also provides services such as authentication and authorization to ensure that only authorized users can access and use the application.

The application layer is the layer closest to the end user and its services are directly accessed by the end user. It works closely with the other layers of the OSI model to ensure that communication between applications is reliable , efficient , and secure.

TCP/IP reference models

Imagine you have a set of instructions to play a game with your friends. To make sure everyone understands each other, you need a common way of communicating. That's where TCP/IP comes in.

TCP/IP is like a special rulebook for communication between computers and devices connected to the internet. It helps them talk to each other and play nicely together.

It consists of the four layers : the application layer , transport layer , internet layer and network access layer.

Application Layer :- The application layer is responsible for providing communication services directly to applications running on different devices. It includes protocols such as HTTP , FTP , SMTP , which are used for services such as web browsing , email and remote login.

Transport Layer :- The transport layer Is responsible for providing reliable data transport services between applications running on different

devices. It includes protocols such as TCP and UDP , which are used for reliable and connection less data transport layer respectively.

Internet Layer :- The internet layer is responsible for routing different networks , it includes the internet protocol (IP) , which is used to provide logical addressing and routing of data packets between devices on different networks.

Network Access Layer: The network access layer is responsible for providing physical access to the network and transmitting data between devices on the same network. It includes protocols such as Ethernet, Wi-Fi, and DSL, which are used for different types of network connections

DIFFERENCE BETWEEN OSI AND TCP/IP

Imagine you want to send a letter to your friend who lives far away. There are certain steps you need to follow to make sure the letter reaches your friend correctly. In computer networks, we use a similar concept called protocols to send information between devices.

The OSI (Open Systems Interconnection) model and the TCP/IP (Transmission Control Protocol/Internet Protocol) model are two different ways of organizing these protocols. They both help us make sure information travels smoothly across networks.

Let's compare them to building a tower with different floors:

1. **OSI Layer:** Think of a tall building with seven floors. Each floor has a specific job to do, and each floor depends on the one below it. The lower floors handle more basic tasks, while the higher floors handle more complex tasks. These floors are called layers. Each layer has a particular responsibility in the communication process.
2. **TCP/IP:** Now, imagine another building with just four floors. The first floor is like the foundation of the building, and the upper floors build upon it. These floors are called protocols. Each floor in this building has multiple functions and

responsibilities, and they work together to make sure the communication works properly.

So, the main difference between the OSI model and the TCP/IP model is the number of layers and how they are organized. The OSI model has more layers, which helps us understand the different tasks involved in communication more precisely. On the other hand, the TCP/IP model has fewer layers, which simplifies the process but still covers all the necessary functions.

Both models help us understand how data moves across a network and ensure that information reaches its destination accurately, just like following steps to send a letter to your friend.

While both models serve the same purpose of organizing protocols for network communication, they were developed by different organizations at different times. The OSI model was created by the International Organization for Standardization (ISO) in the late 1970s, while the TCP/IP model was developed by the U.S. Department of Defense in the 1970s.

APPLICATION LAYER PROTOCOLS

Let's define these commonly used application layer protocols in simpler terms:

1. **File Transfer Protocol (FTP):** Think of FTP as a way to send and receive files over the internet. It's like having a special mailbox where you can put files and someone else can pick them up from there. You can use FTP to share documents, pictures, or any other files with others easily.
2. **Hypertext Transfer Protocol (HTTP):** Whenever you visit a website, your web browser uses HTTP to communicate with the server where the website is stored. It's like a language that allows your browser to ask for web pages and receive them in return. HTTP is responsible for loading and displaying web pages on your computer or device.

3. Simple Mail Transfer Protocol (SMTP):

SMTP is like a postal service for emails. It's the protocol that helps your email get sent from your computer or phone to the recipient's email server. SMTP ensures that your email message is properly addressed, packaged, and delivered to the right inbox so that the recipient can read it.

4. **Domain Name System (DNS):** When you type a website address in your web browser, DNS helps translate that address into the actual numerical address (IP address) of the server where the website is located. It's like looking up a phone number in a directory. DNS helps your computer find the right server so that you can access the website you want.

In simpler terms, these protocols enable you to transfer files, view websites, send emails, and navigate the internet more easily. They provide the necessary rules and methods for different types of communication and interactions on the internet.

TRANSPORT LAYER PROTOCOLS

Let's explain the transport layer protocols, TCP and UDP, in simpler terms:

Transport Control Protocol (TCP): Imagine you want to send a special toy car to your friend who lives far away. TCP is like a reliable delivery service that ensures the toy car reaches your friend safely and in the right order.

TCP breaks down the toy car into small pieces and puts them in separate boxes. It then numbers each box so that your friend can put them back together correctly. TCP also keeps track of the boxes, making sure they all arrive and none of them are lost.

Once all the boxes with the toy car pieces arrive, TCP helps your friend assemble them in the right order, just like putting a puzzle together. This way, your friend receives the complete toy car exactly as you sent it.

User Datagram Protocol (UDP): UDP is like a faster delivery service, but it doesn't offer the same level of reliability as TCP. It's more like sending a postcard to your friend instead of a package.

With UDP, you write your message or draw a picture on a postcard and send it to your friend. However, there is no guarantee that the postcard will arrive or that it will arrive in order. Some postcards might get lost along the way, and your friend may receive them out of order.

UDP is useful when speed is more important than ensuring every piece of information arrives perfectly. It's like sending a quick message to your friend, knowing that if a few postcards go missing or arrive in the wrong order, it's not a big problem.

We call UDP a "connectionless protocol" because it does not establish a dedicated and continuous communication channel between the sender and receiver before sending data.

When you send data using UDP, it does not set up a direct and persistent connection between the sender and receiver. The data is sent as individual packets, like separate messages, without any requirement for establishing or maintaining a connection.

UDP packets are sent to the destination without any confirmation or acknowledgement of receipt. The receiver will either receive the packets or they may be lost in transit. There is no built-in mechanism in UDP to guarantee that all packets arrive, arrive in order, or that there are no duplications.

This connectionless nature of UDP allows for fast and lightweight communication, as there is less overhead involved in establishing and maintaining a connection. However, it also means that UDP does not provide the same level of reliability and error correction as protocols like TCP, which establish a connection and ensure the orderly delivery of data.

So, UDP is called a connectionless protocol because it does not establish a dedicated

connection before sending data and does not require ongoing communication or confirmation of receipt.

In contrast, Transmission Control Protocol (TCP): TCP is like a conversation between two people where they make sure they understand each other by acknowledging and confirming every message they exchange.

Imagine you and your friend are playing a game where you send instructions to build a tower using blocks. With TCP, every time you send an instruction, your friend will confirm that they received it by saying, "Got it!" Then, once they have completed the step, they will let you know by saying, "Done!"

If any instruction or message gets lost along the way, you will ask your friend to repeat it until you both are sure that the message was received and the tower is being built correctly.

TCP ensures that all the information you send, like the instructions for the tower, arrives in the right order and without any missing pieces. It's like having a reliable communication channel where you and your friend double-check everything to make sure there are no mistakes.

In simpler terms, TCP is like a careful and reliable delivery service that ensures all the pieces of a toy car reach your friend and are assembled correctly. UDP is like a faster but less reliable service where you can send messages or pictures quickly, but some pieces might be lost or arrive out of order.

These transport layer protocols help in sending and receiving data over the internet, making sure it arrives accurately (TCP) or quickly (UDP) depending on the needs of the communication.

NETWORK LAYER PROTOCOL

Internet Protocol (IP): IP is the foundational protocol of the internet and is responsible for addressing and routing data packets across networks. It assigns a unique IP address to each device connected to a network and ensures that packets are correctly delivered to their intended destinations.

Imagine you have a bunch of letters that you want to send to your friends who live in different houses. You need a system to ensure that each letter reaches the correct friend's house. That's where the Internet Protocol (IP) comes in.

IP is like a special system that assigns a unique address to each device connected to a network, just like each house has its own address. This address is called an IP address, and it's like the friend's house number.

When you send a letter or any other data over the internet, IP makes sure that it's put in the right "envelope" with the correct IP address of the destination device. It's like writing your friend's address on the envelope so that the postman knows where to deliver the letter.

Once the letter is ready, IP takes care of sending it across different networks to reach the destination. It's like the postman finding the best route to deliver the letter, using roads and intersections.

The job of IP is to make sure that all the letters (data packets) are correctly delivered to the right devices (friends' houses) based on their unique IP addresses. It ensures that the data packets are routed through the internet and reach their intended destinations accurately.

In simpler terms, IP is like a system that gives every device connected to the internet its own special address. It then takes care of putting data in the right envelopes and delivering them to the correct devices, just like sending letters to your friends' houses.

CHAPTER Three

TRANSMISSION MEDIA

Transmission Medium is the physical path between transmitter and receiver in a data communication. The characteristics and quality of data transmission are determined both by the characteristics of the medium and characteristics of the signal.

Medium of data transmission :-

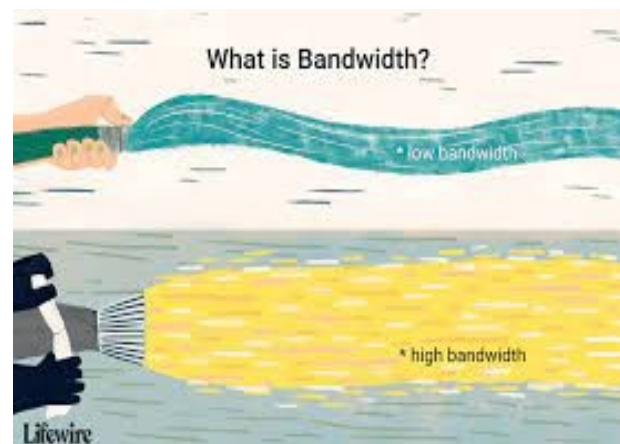
1) Guided Transmission Media :- Data transmission is through solid medium

2) Unguided Transmission Media :- Data transmission is through air (that is wireless system)

GUIDED TRANSMISSION MEDIA :-

Transmission capacity of guided media is dependent on :-

Bandwidth :- Bandwidth is the context of computer networking , **refers to the amount of data that can be transmitted over a network connection in a given amount of time.** Bandwidth is an important consideration in computer networking because it can affect the speed and efficiency on a network. A network with a higher bandwidth will generally be able to transfer data more quickly and efficiently than a network with a lower bandwidth.



Distance :- Distance is an important factor to consider when using guided transmission media , which refers to network transmission media that

use physical cables or wires to transmit data signals. The maximum distance that data can be transmitted over a guided media depends on several factors including :-

- 1) **Type of cable** :- Different types of cables have different transmission distances , for example , twisted pair cables typically have shorter transmission distances than fiber optic cables.
- 2) **Signal strength** :-The strength of the signal being transmitted can affect the distance it can travel. A stronger signal can travel farther than a weaker signal.
- 3) **Attenuation** :- Refers to the loss of signal strength as it travels through cable. Attenuation increases with distance it can travel , so the farther the signal has to travel , the weaker it will become.
- 4) **Interference** :- Interference from other electronic devices or electromagnetic fields can also affect the distance that a signal can travel. The greater the distance , the more likely it is that the signal will be affected by interference.

TYPES OF CONNECTION (POINT TO POINT OR MULTI POINT)

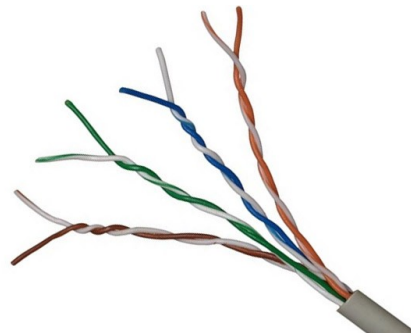
POINT TO POINT CONNECTION :- A point to point connection is a type of network connection in which two devices are directly connected to each other over a dedicated communication channel. This channel can be a physical cable , such as a twisted pair or a fiber optic cable or a wireless link such as a micro-wave link or a satellite link. In a point to point connection , the devices can communicate with each other without the need for any intermediate devices or network components. They can be used in computer networking to connect two devices , such as a computer and a printer or a computer and a server.

Multi point connection :- A multi point is a type of network connection in which multiple devices are connected to a single communication channel. In a multi point connection , the devices share the same communication medium , such as a physical cable or a wireless link.

The three guided media commonly used for data transmission are :-

- a) **Twisted pair**
- b) **Coaxial cable**
- c) **Optical fiber**

A) **TWISTED PAIR** :- is a type of cabling that consists of two copper wires twisted together to reduce electromagnetic interference. It is commonly used in LANs (Local Area Networks) for connecting computers and other network devices. Twisted pair cables come in two varieties : unshielded twisted pair (UTP) and shielded twisted pair (STP). UTP cables are the most common and are used for general-purpose networking , while STP cables are used in environments with high levels of electromagnetic interference.



UNSHIELDED TWISTED PAIR (UTP) CABLE :-

UTP cable is the most common type of twisted pair cable used in networking applications. It consists of two copper wires twisted together , with no additional shielding or insulation. UTP cable is relatively in-expensive and easy to install , and it is suitable for most LAN applications. However , it is more susceptible to electromagnetic interference (EMI) and cross talk than shielded cable.

Advantages of UTP:

1. **Large installed base and familiarity:** UTP has been widely used in networks for many years, so there is a large number of existing installations and expertise available.
2. **Cost-effective and easy installation:** UTP cables are relatively inexpensive compared to other types of network cables. They are

also easy to install, making it convenient for setting up network connections.

3. Compatibility with most LAN systems:

UTP is compatible with most Local Area Network (LAN) systems, which are commonly used for connecting devices within a small area like offices or homes. This means that UTP can work well with many types of devices and network equipment.

such as industrial settings or areas with high electrical activity.



Disadvantages of UTP:

1. **Susceptibility to interference:** UTP cables can be more vulnerable to external **electromagnetic interference, crosstalk (interference between adjacent wires), and signal loss (attenuation) compared to other cable types.** This can affect the quality and reliability of network connections.
2. **Limited security:** UTP may not be the best choice for networks that require a high level of security. Since UTP cables are not shielded, they are more susceptible to eavesdropping or unauthorized access to transmitted data.
3. **Limited suitability for long-distance, high-speed transmission:** UTP may not be suitable for transmitting data over long distances at high speeds. As the length of the cable increases, the quality of the signal may degrade, leading to lower transmission speeds and potential data errors.

SHIELDED TWISTED PAIR (STP) CABLE

STB cable is a type of twisted pair cable that has an additional layer shielding around the individuals pairs of wires. The shielding helps to reduce electromagnetic interference and cross talk, resulting in better signal quality and higher data transmission rates. However, STP cable is more expensive and difficult to install than UTP cable due to the added shielding and insulation. STP cable is commonly used in environments with high levels of electro magnetic interference,

Advantage of STP:

1. **Better performance in noisy environments:** STP cables have a protective metal shield that surrounds each pair of twisted wires. This shield helps to minimize electromagnetic interference (EMI) and radio frequency interference (RFI) from external sources. As a result, STP can provide better signal quality and reliability in environments with high levels of noise.

Disadvantages of STP:

1. **More labor-intensive installation:** Compared to UTP, the installation of STP cables is more labor-intensive. This is because STP cables have an additional metal shield that needs to be properly terminated and grounded. The shielding requires extra care and attention during installation to ensure proper performance.
2. **Compatibility limitations:** Not all LAN systems are readily compatible with STP. Some network devices and equipment may not be designed to work specifically with STP cables. This means that in certain cases, additional configuration or compatibility checks may be required before using STP in a particular network setup.

B) COAXIAL CABLE :-

The term "coaxial" comes from the word "coaxiality," which refers to the arrangement of two or more objects with a common axis. In the case of coaxial cable, it is named so because it consists of two concentric conductors that share the same axis.

The inner conductor and the outer shield of the cable are aligned along the same centerline, or axis, giving it the name "coaxial." The outer shield surrounds the inner conductor and is separated from it by an insulating material. This design helps to maintain the integrity and quality of the electrical signals being transmitted through the cable.



The inner conductor and the outer shield in a coaxial cable serve different roles in the transmission of electrical signals:

1. **Inner Conductor:** The inner conductor, typically made of copper or another conductive material, carries the electrical signal. It serves as the primary path for the transmission of the signal from one end of the cable to the other. The inner conductor is usually a solid wire or a stranded wire, depending on the specific type of coaxial cable.
2. **Outer Shield:** The outer shield, which is usually made of metal such as aluminum or copper, surrounds the inner conductor. Its primary function is to provide a layer of protection and minimize interference or

noise from external electromagnetic sources. The shield acts as a barrier, preventing unwanted signals from entering the cable and interfering with the transmitted signal. It also helps contain and minimize any electromagnetic radiation emitted by the inner conductor, reducing the potential for signal leakage.

Together, the inner conductor and outer shield work in tandem to ensure the reliable transmission of electrical signals. The inner conductor carries the desired signal, while the outer shield helps protect it from external interference and preserves signal integrity. This makes coaxial cable suitable for applications that require high-quality signal transmission, such as in television broadcasting, networking, and cable television systems.

Advantages of Coaxial Cable:

1. **Less Susceptible to Interference:** Coaxial cable is better at resisting interference from external sources compared to twisted-pair cables. This means that the signals transmitted through coaxial cable are less likely to be distorted or weakened by outside electrical signals.
2. **Longer Transmission Distance:** Coaxial cable can transmit data over longer distances compared to less expensive cables. This makes it useful for applications where data needs to be transmitted over a significant distance without loss of quality or signal strength.
3. **Supports Voice, Video, and Data:** Coaxial cable is capable of transmitting different types of signals, including voice, video, and data. This versatility makes it suitable for applications such as television broadcasting, internet connectivity, and telephone services.
4. **Reasonable Data Security:** Coaxial cable offers a certain level of data security. Its design makes it difficult for unauthorized parties to tap into the cable and access the transmitted information.

Disadvantages of Coaxial Cable:

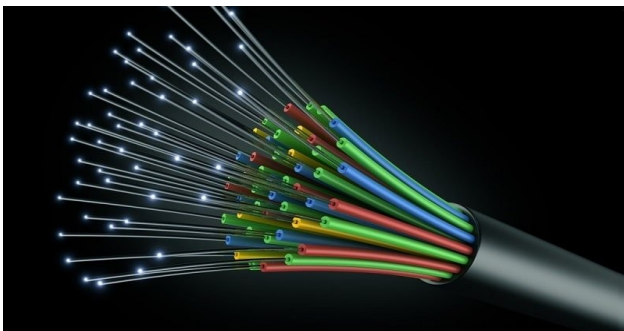
1. **Limited Compatibility:** There are various types of coaxial cables, and each type is suited for specific types of local area network (LAN) systems. This means that not all LAN systems can readily work with any type of coaxial cable. It may require careful selection and compatibility considerations.
2. **Higher Cost:** Coaxial cable tends to be more expensive compared to other cable types. This is partly due to its higher metallic content and the specialized manufacturing processes involved in its production.

In simpler terms, coaxial cable has some advantages over other cables. It is less likely to be affected by outside interference, can transmit data over long distances, and supports different types of signals. However, there are different types of coaxial cables, and they may not be compatible with all LAN systems. Additionally, coaxial cable can be more expensive than other cables due to its construction and materials.

C) Fiber Optics

Optical fiber cable is a type of cable that is used to transmit information in the form of light signals. It is made up of a very thin and flexible strand called an optical fiber. This fiber is usually made of glass or plastic and is incredibly thin, even thinner than a single strand of human hair.

The light signals that travel through the optical fiber carry information in the form of binary code, which is a series of ones and zeros. These signals can represent various types of data, such as voice, video, or internet data.



The main reason why optical fiber cable is preferred over other methods like coaxial and twisted pair cable is because of its superior performance and advantages:

1. **High Speed and Bandwidth:** Optical fiber cable can transmit data at very high speeds, allowing for fast and efficient communication. It has a much higher bandwidth compared to coaxial and twisted pair cables, meaning it can carry a larger amount of data simultaneously.
2. **Long Transmission Distance:** Optical fiber cable can transmit data over very long distances without significant loss of signal quality. It can span hundreds of kilometers without the need for signal boosters or repeaters, making it suitable for long-distance communication.
3. **Immunity to Electromagnetic Interference:** Unlike coaxial and twisted pair cables, optical fiber cable is not affected by electromagnetic interference. This means it can transmit data without being affected by nearby power lines, electronic devices, or other sources of electromagnetic radiation. It provides a stable and reliable connection.
4. **Security:** Optical fiber cable offers a high level of security for data transmission. Since it uses light signals to transmit data, it is very difficult to tap into the cable and intercept the information being transmitted. This makes it ideal for applications that require secure communication, such as financial transactions or sensitive data transfer.
5. **Lightweight and Space Efficient:** Optical fiber cable is lightweight and takes up less space compared to other cables. It can be easily installed and doesn't require as much physical space. This makes it suitable for environments where space is limited, such as in crowded data centers or underground installations.

6. **Future-Proof Technology:** Optical fiber cable is considered a future-proof technology, meaning it is designed to meet the demands of future communication needs. It has the potential to support even higher data rates and can easily be upgraded as technology advances.

Optical fiber cable is preferred over coaxial and twisted pair cables because it offers higher speeds, longer transmission distances, immunity to interference, enhanced security, and is more space-efficient. These advantages make it the preferred choice for high-speed internet, telecommunication networks, and data transmission in various industries.

While optical fiber cables offer numerous advantages, they also have some disadvantages to consider:

1. **Cost:** The initial installation cost of optical fiber cables can be higher compared to copper-based systems. This is primarily due to the higher cost of the fiber optic cable itself, as well as the specialized equipment and electronics required for transmitting and receiving light signals. However, the long-term cost savings of optical fiber cables, such as lower maintenance and higher data transmission capacity, can outweigh the initial investment.
2. **Electronics and Equipment:** Optical fiber cables require specialized electronics for converting light signals to electrical signals and vice versa. These components, such as transmitters and receivers, can be more expensive than those used in copper-based systems. Additionally, the maintenance and replacement of these electronics can also add to the overall cost.
3. **Specialized Installation:** Installing optical fiber cables requires specialized knowledge and expertise. The cables need to be carefully handled and protected from

bending or breaking, as any damage to the fiber can lead to signal loss or degradation. **Specialized tools and techniques are necessary to splice and connect the fibers during installation.** This means that the installation process may be more time-consuming and require trained professionals.

4. **Limited Flexibility:** Optical fiber cables are not as flexible as other types of cables, such as twisted pair or coaxial cables. They are more delicate and can be more challenging to bend around tight corners or in small spaces. This can make installation in certain environments or structures more difficult.
5. **Compatibility:** Optical fiber cables are not directly compatible with existing copper-based infrastructure. In some cases, organizations may need to invest in additional equipment, such as media converters or optical network interfaces, to bridge the gap between fiber optics and traditional copper-based systems. This compatibility issue can add complexity and cost to the network setup.

It's important to note that while these disadvantages exist, the benefits of optical fiber cables often outweigh them in many applications where high-speed, long-distance, and secure data transmission is essential. The disadvantages can be mitigated with proper planning, expertise, and consideration of the specific requirements of the network infrastructure.

UNGUIDED TRANSMISSION MEDIUM (WIRELESS TRANSMISSION)

Unguided transmission media, also known as wireless or wireless communication, refers to a type of transmission medium that does not require physical cables or wires to transmit signals. Instead, it utilizes various methods such as radio waves, microwaves, or infrared signals to carry information through the air or free space.

In the context of unguided transmission media, directional and omnidirectional refer to the patterns in which the wireless signals are transmitted or received.

1. Directional-Transmission/Reception:

Directional transmission or reception involves focusing the wireless signals in a specific direction or within a narrow beam. This means that the signals are concentrated and intended to travel in a particular path or towards a specific target. Directional transmission/reception is achieved using specialized antennas, such as parabolic or dish antennas, which can concentrate the signal energy in a specific direction. This approach is useful for long-range communication, point-to-point links, and scenarios where the signal needs to be focused in a particular direction to reach a specific receiver.

2. Omnidirectional Transmission/Reception:

Omnidirectional transmission or reception involves spreading the wireless signals in all directions, creating a 360-degree coverage pattern. With omnidirectional transmission/reception, the signals radiate outwards in all directions from the source or antenna. This allows the signals to reach multiple devices or receivers located in different directions without the need for precise alignment. Omnidirectional antennas, such as dipole or whip antennas, are commonly used to achieve this coverage pattern. This approach is suitable for scenarios where the signal needs to reach multiple devices in different locations, such as in Wi-Fi networks or cellular networks where devices are spread throughout an area.



Here are some simple points about radio waves:

1. **Invisible Waves:** Radio waves are like invisible ripples that travel through the air. We can't see them, but they are all around us, helping us communicate wirelessly.
2. **Sending Messages:** Just like how we can talk to someone far away using a telephone, radio waves help us send messages wirelessly. We can use them to listen to the radio, watch TV, talk on a cell phone, or connect to the internet using Wi-Fi.
3. **Long Distance Travel:** Radio waves are great at traveling long distances. They can go through the air and even outer space to reach faraway places. That's why we can listen to radio stations or make phone calls to people who are far from us.
4. **Antennas:** To send and receive radio waves, we use special antennas. Antennas are like magic sticks that can catch and send out these invisible waves. They are the reason we can listen to music on the radio or watch TV shows on our antennas at home.
5. **Different Channels:** Just like we tune our radio or TV to different channels, radio waves have different "channels" too. Each channel carries different types of signals, like music, news, or TV shows. That's why we can listen to different stations or watch different channels on TV.
6. **Interference:** Sometimes, there can be too many signals in the air, and they can mix

TYPES OF WIRELESS NETWORKS

Radio Waves :- Imagine radio waves as invisible waves in the air that can carry messages from one place to another without using any wires. They are a special kind of energy called electromagnetic radiation.

up or interfere with each other. It's like trying to hear one person talking in a crowded room. That's why we might sometimes experience static or poor reception when listening to the radio or watching TV.

Remember, radio waves are like magical invisible waves in the air that help us stay connected wirelessly. They bring us music, TV shows, phone calls, and so much more, making our lives more fun and convenient!

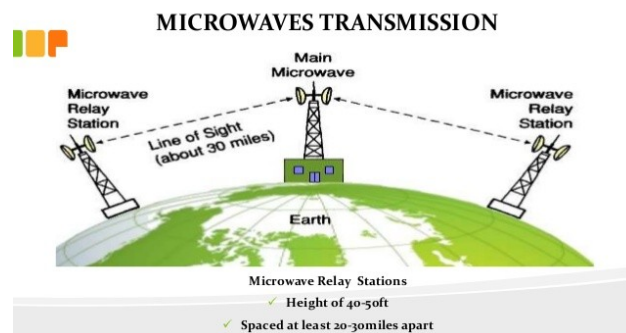
Micro Waves

Microwaves are a type of invisible waves that are used for wireless communication, just like radio waves. They are called microwaves because they have a very short wavelength, which means they are really tiny and close together.

Here are some simple points about microwaves:

1. **Tiny Waves:** Microwaves are like tiny waves that travel through the air. They are smaller than radio waves and can fit many of them in a small space. Just like how you can see lots of little ripples in a pond, microwaves are tiny ripples in the air.
2. **Fast Communication:** Microwaves are really good at carrying messages and information quickly. They are often used for things like cell phones, satellite communication, and Wi-Fi. They can send signals from one place to another in the blink of an eye!
3. **Cooking with Microwaves:** You may have heard of microwaves being used in microwave ovens. These special ovens use microwaves to heat up and cook our food. Microwaves can make the water molecules in the food vibrate, which creates heat and cooks the food.
4. **Line of Sight:** Unlike radio waves, microwaves need a clear path to travel. They like to move in straight lines and can be blocked by obstacles like buildings

or mountains. That's why you might notice that your Wi-Fi signal gets weaker if you move to a different room.



5. **High Frequencies:** Microwaves have higher frequencies than radio waves, which means they vibrate really fast. This high frequency allows them to carry a lot of information and data, making them great for fast internet connections.

Applications: Radio waves are commonly used for broadcasting radio and television signals, as well as for communication in AM/FM radio, shortwave radio, and satellite communication. Microwaves, on the other hand, are utilized in various technologies, such as microwave ovens, Wi-Fi networks, cellular communication (e.g., 4G and 5G), radar systems, and satellite communication.

Infrared

Imagine you have a special kind of light that you can't see with your eyes, but you can feel it as heat. That's what infrared waves are. They are a type of energy that is just below the red color that we can see.

Infrared waves have shorter wavelengths than radio waves and microwaves, but longer wavelengths than visible light. This means they have less energy than visible light, but more energy than radio waves and microwaves.

Infrared waves are interesting because they can help us in many ways. Here are a few examples:

1. **Remote Controls:** Have you ever used a remote control to change the TV channel or adjust the volume? Well, remote controls use infrared waves to send

signals to the TV. When you press a button on the remote, it sends out a beam of infrared waves that the TV can understand.



2. **Heat Sensing:** Infrared waves can also help us detect heat. Special cameras called infrared cameras can see things that give off heat, even in the dark. This can be useful for firefighters to find people inside a smokey building or for scientists to study how animals stay warm.



3. **Cooking:** Infrared waves are also used in cooking. You know when you use a toaster or a grill to cook your food? They use infrared waves to heat up the food. These waves can penetrate the food and make it cook faster and more evenly.

Infrared waves are a type of energy that we can feel as heat but can't see with our eyes. They have shorter wavelengths than radio waves and microwaves, but longer wavelengths than visible light. Infrared waves have many practical applications, such as remote controls, heat sensing, and cooking.

CHAPTER FOUR

Network Layer Addressing and Routing

Imagine you have a big school with lots of classrooms and students. Each student has their own unique student ID number. In a similar way, every device (like computers, smartphones, or tablets) connected to a computer network also needs a unique identification number to communicate with other devices. This identification number is called an **IP address**.

Network Layer Addressing :- The network layer addressing is like giving each device in the network its own address. Just like you have a home address, each device has its own IP address. An IP address is a combination of numbers (like 192.168.1.1) that helps in identifying a specific device on the network.

Routing: Now, imagine you want to send a message to your friend who is in a different classroom. You need to know how to get the message to your friend's classroom. In a computer network, routing is the process of figuring out the best path or route for your message to reach its destination.

To do this, there are special devices called **routers** that help in routing messages. Routers are like the school's staff members who know the layout of the school and the best way to reach different classrooms. They receive messages and decide where to send them based on the IP addresses of the sender and recipient.

When you send a message, the router checks the recipient's IP address and consults its routing table to determine the next router or path the message should take. Each router looks at the destination IP address and decides which way the message should go until it reaches the correct device.

The routers work together like a team, passing the message from one router to another until it reaches its destination. They make sure the message follows the correct path, just like a teacher guiding you to the right classroom.

So, network layer addressing gives each device its own IP address, and routing ensures that messages are correctly directed to their destination based on those addresses.

Network Routing

Imagine you have a large city with many houses, and each house has a unique address. In a similar way, devices connected to a computer network, like computers or smartphones, also need unique identification numbers to communicate with each other. These identification numbers are called IP addresses.

Network Routing: Now, think of a scenario where you want to send a letter from your house to your friend's house on the other side of the city. You know the addresses of both houses, but you need to figure out the best route to deliver the letter. This process of finding the best route for your letter to reach its destination is called routing.

Routing is done by special devices called **routers**. Routers are like post offices in our city. They receive the letter and decide where to send it based on the address. Routers have a table that helps them determine the next hop, or the next router, the letter should be forwarded to.

Types of Routing:

1. **Unicast Routing:** When you send a letter to a specific friend, it's like unicast routing. You know the exact address of your friend's house, so you simply write the address on the envelope and give it to the post office. The router looks at the destination address and forwards the letter to the next router or directly to your friend's device.

Unicast routing is the most common type of routing used on the internet and intranets. It involves sending data to a specific destination. When a device wants to send a packet to a particular destination, it looks up the routing table on a router to determine the next hop or

the next router that the packet should be forwarded to. The process continues until the packet reaches its intended destination.

2. **Broadcast Routing:** Sometimes, you may want to send a message to everyone in your neighborhood. In that case, you write the message on a poster and give it to the post office. The post office will make copies of the poster and deliver them to every house in the neighborhood. This is similar to broadcast routing, where a message is sent to all devices on a network.

By default, routers do not forward broadcast packets. Broadcast packets are messages sent to all devices on a network. Routers create separate broadcast domains, meaning that a broadcast packet is limited to the devices within that domain. However, in certain cases, routers can be configured to forward broadcasts. There are two methods for broadcast routing:

- The first method involves the router creating multiple copies of a single data packet, each with a different destination address. The router sends these packets to each host individually as a unicast message, creating the illusion of broadcasting.
 - The second method, known as flooding, involves the router forwarding the broadcast packets out of all its interfaces to reach all devices within the broadcast domain. All routers in the network are configured in the same way to ensure the broadcast packets are delivered.
3. **Multicast Routing:** Let's say you want to invite only a specific group of friends to a party. You create a list of friends who should receive the invitation and give it to the post office. The post office delivers the invitation only to the friends on the list. This is like multicast routing, where data is sent only to a specific group of

devices that have expressed interest in receiving the data.

Multicast routing is a special case of broadcast routing, but with a significant difference. With multicast routing, data is sent only to those devices that have expressed a desire to receive the multicast packets. This is different from broadcast routing, where packets are sent to all devices regardless of their interest. Multicast routing utilizes the spanning tree protocol to prevent loops in the network. It also employs the reverse path forwarding technique to detect and discard duplicate packets and avoid looping.

4. **Anycast Routing:** Anycast routing is a bit different. Imagine you have multiple pizza places in the city, and each place offers the same type of pizza. When someone orders a pizza, the order is automatically routed to the nearest pizza place. This is similar to anycast routing. Multiple devices share the same address, but the data is sent to the nearest device among them based on the routing topology.

Anycast routing allows multiple hosts to share the same logical address. When a packet is sent to this shared address, it is forwarded to the host that is nearest in terms of routing topology.

These different types of routing provide mechanisms for efficiently and effectively delivering data within networks based on specific requirements. Unicast routing is used for sending data to a specific destination, broadcast routing is used for sending data to all devices within a network, multicast routing is used for sending data to a specific group of interested devices, and anycast routing is used for directing data to the nearest host among multiple hosts sharing the same logical address.

IP - ADDRESS

INTERNET PROTOCOL VERSION 4

IPv4, or Internet Protocol Version 4, is a fundamental communication protocol used in computer networks, including the internet. It is the fourth version of the Internet Protocol and has been widely used since its development in the 1980s. Let's dive into a detailed explanation of IPv4:

1. **What is IPv4?** IPv4 is a system that provides a way for devices to communicate and exchange data over the internet. It assigns a unique numerical address to each device connected to a network. These addresses are called IP addresses and serve as identification numbers for devices.
2. **Why do we need IPv4?** The main purpose of IPv4 is to enable communication between different devices on a network, such as computers, servers, routers, and smartphones. IP addresses allow devices to find and send data to one another across the internet. IPv4 is a foundational technology that allows us to browse websites, send emails, stream videos, and perform various online activities.
3. **Why is it called "Version 4"?** IPv4 is referred to as "Version 4" because it is the fourth iteration of the Internet Protocol. The previous versions, IPv1, IPv2, and IPv3, were experimental and not widely adopted. IPv4 was developed as the first version of IP intended for global deployment and widespread use. Its design and specifications were refined to become the most widely used version of IP at that time.
4. **Addressing Scheme and Structure:** IPv4 addresses are 32 bits long, represented as four sets of numbers ranging from 0 to 255, separated by periods. For example, an IPv4 address can look like 192.168.0.1.

These addresses are divided into different classes:

IPv4 addresses are divided into different classes, which determine how the address is structured and how it is used. The classes are as follows:

1. Class A:

- In Class A addresses, the first octet (the first set of numbers) is used to identify the network, while the remaining three octets represent the host address.
- The range of the first octet in Class A addresses is from 1 to 126.
- This class is used for large networks, such as multinational corporations or internet service providers (ISPs).
- Class A addresses can support a large number of hosts, as the host address portion has 24 bits (three octets).

2. Class B:

- In Class B addresses, the first two octets are used for network identification, and the other two octets indicate the host address.
- The range of the first octet in Class B addresses is from 128 to 191.
- Class B addresses are typically used for medium-sized networks, such as universities or large organizations.
- Class B addresses can support a moderate number of hosts, as the host address portion has 16 bits (two octets).

3. Class C:

- In Class C addresses, the first three octets identify the network, while the last octet specifies the host address.
- The range of the first octet in Class C addresses is from 192 to 223.
- Class C addresses are commonly used for small networks, such as small businesses or home networks.
- Class C addresses can support a limited number of hosts, as the host address portion has 8 bits (one octet).

Each class of address has a specific range reserved for it and follows a defined structure, making it easier to allocate and manage IP addresses across different networks.

Understanding the classes of IPv4 addresses helps in organizing and identifying networks based on their size and requirements. It ensures that IP addresses are used efficiently and allows devices within a network to communicate with each other accurately.

Address Exhaustion: Address exhaustion refers to the situation where the available pool of unique IP addresses within the IPv4 system becomes depleted. In the early days of the internet, when IPv4 was developed, it was anticipated that the 32-bit addressing scheme (providing around 4.3 billion addresses) would be sufficient. However, with the rapid growth of the internet and the increasing number of devices connecting to it, the demand for IP addresses exceeded the available supply.

Transition to IPv6: To overcome the limitation of available addresses in IPv4, a newer version of the Internet Protocol called IPv6 (Internet Protocol Version 6) was developed. IPv6 was designed to provide a significantly larger address space to meet the growing demands of internet-connected devices.

Here are some key aspects of the transition to IPv6:

1. **Address Space:** IPv6 uses 128-bit addresses, which provide an incredibly vast number of unique addresses compared to the 32-bit addresses used in IPv4. The larger address space in IPv6 allows for trillions of devices to be connected to the internet, ensuring an abundant supply of addresses for future growth.
2. **Address Format:** IPv6 addresses are represented in a different format compared to IPv4. Instead of four sets of numbers separated by periods, IPv6 addresses are expressed as eight groups of four hexadecimal digits separated by

colons. For example, an IPv6 address can look-like

2001:0db8:85a3:0000:0000:8a2e:0370:7334.

3. **Additional Features:** IPv6 introduces various additional features and improvements. One notable feature is built-in support for security and encryption through IPsec (Internet Protocol Security). IPv6 also simplifies address assignment through automatic address configuration, reducing the reliance on servers for assigning addresses. Furthermore, IPv6 includes enhancements for better routing and network management.

The transition to IPv6 is an ongoing process as the internet gradually adopts the new protocol. It aims to address the issue of address exhaustion and ensure a sustainable, long-term solution for the growing number of devices connected to the internet.

Internet Protocol Version 6 (IPv6)

Internet Protocol Version 6 (IPv6) is a newer and improved version of the Internet Protocol, which is like a special system that allows devices to communicate with each other on the internet. Here's what you need to know about IPv6:

1. **More Addresses:** IPv6 was created because we were running out of addresses with the older version, IPv4. Think of addresses like phone numbers for devices. IPv6 provides a much larger number of unique addresses than IPv4, so we can connect many more devices to the internet. It's like having a bigger phone book to accommodate everyone.
2. **Longer Address Format:** In IPv6, addresses are longer and look different compared to IPv4. Instead of four sets of numbers separated by dots, IPv6 addresses are made up of eight groups of numbers and letters separated by colons. It's like having a longer code or a longer secret password for devices to identify and talk to each other.
3. **New Features:** IPv6 brings some new and improved features. One important feature is better security. It helps protect the communication between devices and keeps data safe. It's like having a special lock on your phone or computer to prevent others from accessing your information.
4. **Auto-Configuration:** IPv6 has a cool feature called auto-configuration. This means that devices can give themselves an IPv6 address without needing anyone's help. It's like having a magic ability to know your own special number without asking anyone else. This makes it easier to connect devices to the internet and start using them right away.
5. **Future-Proofing:** IPv6 was designed to handle the growing number of devices that will connect to the internet in the future. It provides a really big address space, so we won't run out of addresses anytime soon. It's like having a super long phone book that can hold everyone's

Imagine you have a special device that can connect to the internet, like a tablet or a computer. When you want to connect it to the internet, you usually need to ask for help to get an address, just like getting your own special number to join a club. But with IPv6, devices have a cool magic power called auto-configuration. It means that the device can figure out its own special address without asking anyone for help. It's like the device can look in a magic mirror and know what its special number is.

- This magic power makes it really easy to connect devices to the internet quickly. It saves time because the device can start using the internet right away without needing someone else to set it up. So, auto-configuration is like having a superpower that allows devices to give themselves a special internet address, like their own secret number, without needing anyone else's help.

5. **Future-Proofing:** IPv6 was designed to handle the growing number of devices that will connect to the internet in the future. It provides a really big address space, so we won't run out of addresses anytime soon. It's like having a super long phone book that can hold everyone's

phone number, even as more people and devices join the internet.

6. **Transition:** Right now, both IPv4 and IPv6 are used together because not everyone has switched to IPv6 yet. It's like having two different languages spoken on the internet. But eventually, as more and more devices use IPv6, it will become the main way of connecting to the internet.

IPv6 is a newer and better version of the Internet Protocol that allows more devices to connect to the internet. It has longer addresses, improved security, and features like auto-configuration. IPv6 was created to handle the future growth of the internet and will eventually replace the older IPv4 system.

INTERNET PROTOCOL VERSION 4 (IP-4) IN DETAIL

An IP address is like a special code that uniquely identifies a device (such as a computer or a router) on the internet. It's similar to a phone number or an address that helps us find a specific person or place.

- **IPv4:** IPv4 is a type of IP address that uses 32 bits to create a unique address. The address is a series of 32 0s and 1s, and it looks something like this: 11001010.01101100.00010000.00000001. Each set of 8 bits is called an octet. The purpose of an IPv4 address is to define a device's connection to the internet.
- **Uniqueness:** IPv4 addresses are unique, meaning that no two devices can have the exact same address at the same time. It's like each house having a different address, so the mail can be delivered to the right place. However, an address can be assigned to one device for a period of time and then taken away and given to another device.

- **Multiple Connections:** If a device, like a router, has multiple connections to the internet, it needs multiple IP addresses to uniquely identify each connection.
- **Universality:** IPv4 addresses are universal, which means that any device wanting to connect to the internet must follow the IPv4 addressing system.
- **Address Space:** The address space refers to the total number of possible addresses that can be used within a specific addressing scheme.
- **IPv4 Address Space:** IPv4 uses 32 bits to define its addresses, which gives a total address space of 2^{32} (2 to the power of 32), resulting in more than 4 billion unique addresses. It's like having a huge room with 4 billion chairs, and each chair represents a unique address. However, due to various restrictions and allocations, not all addresses are available for general use.
- **Theoretical Device Limit:** The vast number of available addresses theoretically allows for more than 4 billion devices to be connected to the internet. However, in reality, the available pool of IPv4 addresses has become limited due to the increasing number of devices connected to the internet.

Notations of IPV4

IPv4 address can be represented in two notations: binary and dotted-decimal. Binary notation shows the address as a sequence of 32 bits, while dotted-decimal notation represents the address as four decimal numbers separated by dots. Converting between the notations involves grouping bits into octets and converting between decimal and binary representations.

Binary Notation:

- In binary notation, an IPv4 address is shown as a sequence of 32 bits, which are either 0s or 1s. These 32 bits are

divided into four groups, each consisting of 8 bits, called octets or bytes.

- Each bit can be thought of as a switch that is either on (1) or off (0).
- For example, let's take the binary IPv4 address "01110101 10010101 00011101 00000010". It is divided into four octets: 01110101, 10010101, 00011101, and 00000010.
- Each octet represents a number from 0 to 255. It's like having four switches that can be turned on or off independently, and each switch represents a number from 0 to 255.

Dotted-Decimal Notation:

- Dotted-decimal notation is a more human-friendly way of representing IPv4 addresses.
- In this notation, the IPv4 address is divided into four parts, or octets, separated by dots. Each octet is represented by a decimal number ranging from 0 to 255.
- For example, let's take the dotted-decimal IPv4 address "117.149.29.2". It consists of four decimal numbers: 117, 149, 29, and 2.
- Each decimal number represents the value of the corresponding octet. It's like having four numbers written side by side, with dots separating them.

Converting between Binary and Dotted-Decimal Notations:

- To convert an IPv4 address from binary to dotted-decimal notation, we group the 32 bits into four octets and convert each octet from binary to decimal. We then separate the decimal numbers with dots.
- To convert from dotted-decimal to binary notation, we take each decimal number and convert it from decimal to binary. We then combine the binary values of the four octets.

By using these notations, we can represent IPv4 addresses in different formats depending on the

purpose or readability required. Binary notation shows the address as a sequence of 32 bits, while dotted-decimal notation represents the address as four decimal numbers separated by dots. Converting between the notations involves grouping the bits into octets and converting between decimal and binary representations.

2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
128	64	32	16	8	4	2	1

Example 1: Binary to Dotted-Decimal Conversion

Let's convert the binary IPv4 address "11000000 10101000 00000001 00000010" to dotted-decimal notation:

Step 1: Group the binary bits into four octets: 11000000 10101000 00000001 00000010

Step 2: Convert each octet from binary to decimal:

$$\times \quad 11000000 = 192$$

$$\times \quad 10101000 = 168$$

$$\times \quad 00000001 = 1$$

$$\times \quad 00000010 = 2$$

Step 3: Combine the decimal values with dots: 192.168.1.2

So, the dotted-decimal representation of the binary IPv4 address "11000000 10101000 00000001 00000010" is 192.168.1.2.

Example 2: Dotted-Decimal to Binary Conversion

Now, let's convert the dotted-decimal IPv4 address "172.16.0.1" to binary notation:

Step 1: Separate the decimal numbers: 172 16 0 1

Step 2: Convert each decimal number to binary: 172 = 10101100 16 = 00010000 0 = 00000000 1 = 00000001

Step 3: Combine the binary values of the octets: 10101100 00010000 00000000 00000001

So, the binary representation of the dotted-decimal IPv4 address "172.16.0.1" is 10101100 00010000 00000000 00000001.

Valid and Invalid IPv4 Addresses

Valid IPv4 Address: An IPv4 address is considered valid if it follows the specific rules and format defined for IPv4 addresses. Here are the key characteristics of a valid IPv4 address :-

1. **Consists of four octets:** An IPv4 address is divided into four parts called octets or bytes. Each octet is represented by a decimal number ranging from 0 to 255. The octets are separated by dots in the dotted-decimal notation.
2. **Numeric values within the valid range:** Each octet of an IPv4 address must have a numeric value between 0 and 255, inclusive. This means that each part of the address can be any number from 0 to 255.
3. **No leading zeros:** Leading zeros (zeros at the beginning of an octet) are not allowed in IPv4 addresses. For example, "192.168.001.001" is not a valid representation because it has leading zeros.

Examples of Valid IPv4 Addresses:

- 192.168.0.1
- 10.0.0.255
- 172.16.254.1

Invalid IPv4 Address: An IPv4 address is considered invalid if it does not meet the requirements and rules for valid IPv4 addresses. Here are some common reasons for an IPv4 address to be invalid:

1. **Incorrect format:** An IPv4 address must consist of four octets separated by dots. If the address has more or fewer than four octets or if the octets are not separated by dots, it is considered invalid.

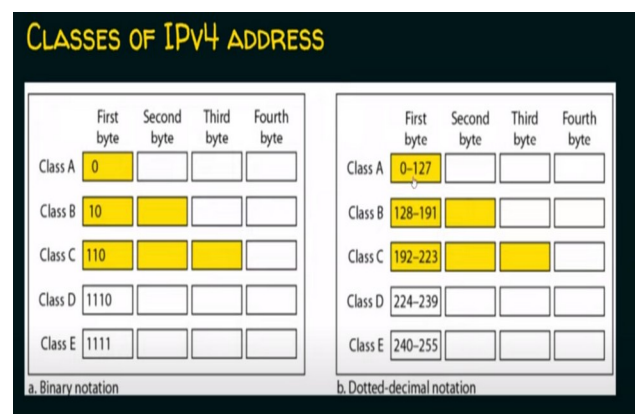
2. **Out of range values:** Each octet of an IPv4 address must have a numeric value between 0 and 255. If any part of the address has a value outside this range, the address is invalid.
3. **Non-numeric characters:** IPv4 addresses should only consist of numeric values and dots. If an address contains non-numeric characters or special symbols (except dots), it is considered invalid.
4. **Leading or trailing dots:** An IPv4 address should not have leading or trailing dots. For example, ".192.168.0.1" or "192.168.0.1." are invalid representations.

Examples of Invalid IPv4 Addresses:

- 192.168.0 (missing the last octet)
- 10.256.0.1 (invalid value in octet)
- 192.168.0.01 (leading zero in an octet)
- 192.168.0.1.1 (extra octet)

Remember, a valid IPv4 address follows the defined format, has numeric values within the valid range for each octet, and does not contain any invalid characters. An invalid IPv4 address fails to meet one or more of these criteria.

CLASS FULL IP ADDRESSING



Classful IP addressing refers to the original method of allocating IP addresses in the early days of the internet. In classful addressing, IP addresses were divided into different classes (Class A, Class B, and Class C) based on the size of the network they were assigned to. Each class had a fixed portion of the IP address reserved for

network identification, and the remaining portion was used for host identification.

Here is a detailed explanation of classful IP addressing :-

Class A Addressing :-

- Class A addresses were assigned to large networks, such as multinational corporations or internet service providers (ISPs).
- In a Class A address, the first octet (the first set of numbers) is used to identify the network, and the remaining three octets represent the host address.
- The range of the first octet in Class A addresses is from 0 to 127.
- Class A addresses can support a large number of hosts because the host address portion has 24 bits (three octets).
- Example: 10.0.0.1, where 10 is the network address, and 0.0.1 is the host address.

Class B Addressing :-

- Class B addresses were allocated to medium-sized networks, such as universities or large organizations.
- In a Class B address, the first two octets are used for network identification, and the other two octets indicate the host address.
- The range of the first octet in Class B addresses is from 128 to 191.
- Class B addresses can support a moderate number of hosts because the host address portion has 16 bits (two octets).
- Example: 172.16.0.1, where 172.16 is the network address, and 0.1 is the host address.

Class C Addressing:

- Class C addresses were reserved for small networks, such as small businesses or home networks.
- In a Class C address, the first three octets identify the network, while the last octet specifies the host address.

- The range of the first octet in Class C addresses is from 192 to 223.
- Class C addresses can support a limited number of hosts because the host address portion has 8 bits (one octet).
- Example: 192.168.0.1, where 192.168.0 is the network address, and 1 is the host address.

4. Class D and Class E:

- Class D addresses (starting from 224.0.0.0 to 239.255.255.255) are reserved for multicast addressing, which allows a single packet to be sent to multiple destinations simultaneously.
- Class E addresses (starting from 240.0.0.0 to 255.255.255.255) are reserved for experimental or special purposes.

It's important to note that classful addressing was later replaced by Classless Inter-Domain Routing (CIDR) due to the limitations and inefficiencies of the classful system. CIDR allows for more flexible allocation of IP addresses by using variable-length subnet masks (VLSM) and route aggregation.

Classful IP addressing involved dividing IP addresses into different classes (Class A, B, C) based on the size of the network. Each class had a fixed portion for network identification and a varying portion for host identification. Classful addressing was eventually replaced by CIDR for better address allocation and routing efficiency.

WHY DO WE NEED CLASSES FROM THE FIRST PLACE ?

The concept of classes in IP addressing was introduced in the early days of the internet as a way to allocate IP addresses more efficiently. Here are some reasons why classes were initially used:

1. **Address Space Management:** Classful addressing provided a way to divide the available IP address space into distinct blocks or classes. This allowed for easier management and allocation of addresses

to different organizations, networks, and devices.

2. **Network Size Differentiation:** The classes (Class A, B, and C) were designed to cater to networks of different sizes. By assigning IP addresses based on classes, it was easier to determine the approximate number of hosts that could be accommodated in a particular network.
3. **Routing Efficiency:** Classful addressing helped in simplifying the routing process by allowing routers to determine the network boundaries based on the class of the IP address. This allowed routers to make routing decisions more quickly and efficiently.
4. **Address Assignment Policies:** Each class had specific address assignment policies. For example, Class A addresses were allocated to large organizations, Class B addresses to medium-sized networks, and Class C addresses to small networks. These policies helped ensure that address blocks were assigned according to the size and requirements of the network.

However, over time, it became evident that classful addressing had limitations and inefficiencies. **The main drawback was the rigid allocation of address space, resulting in wastage of addresses and inefficient use of available resources. This led to the introduction of Classless Inter-Domain Routing (CIDR), which allowed for more flexible allocation of addresses using variable-length subnet masks (VLSM).**

CIDR replaced the classful system and provided a more scalable and efficient approach to IP addressing. With CIDR, address blocks could be

divided into smaller subnets, allowing for finer control over address allocation and improved routing efficiency.

What is a Default Mask (Subnet Mask)

A default mask, also known as a subnet mask or network mask, is a fundamental component in IP addressing. It determines how the IP address is divided into network and host portions. The default mask is used to identify which bits in an IP address represent the network and which bits represent the host.

In classful IP addressing, each class (Class A, B, or C) had a predefined default mask associated with it. Here's a description of the default masks for each class :-

Class A Default Mask:

- Class A addresses are designed for large networks.
- The default mask for Class A addresses is 255.0.0.0 or /8 in CIDR notation.
- This default mask means that the first octet represents the network portion of the address, and the remaining three octets represent the host portion.
- Example: If an IP address is 10.0.0.1 with the default mask 255.0.0.0, the network portion is 10, and the host portion is 0.0.1.

Class B Default Mask:

- Class B addresses are meant for medium-sized networks.
- The default mask for Class B addresses is 255.255.0.0 or /16 in CIDR notation.
- This default mask means that the first two octets represent the network portion, and the remaining two octets represent the host portion.
- **Example :-** If an IP address is 172.16.0.1 with the default mask 255.255.0.0, the network portion is 172.16, and the host portion is 0.1.

3. Class C Default Mask:

Address Class	1st Octet range in decimal	1st Octet bits (Blue Dots do not change)	Network (N) and Host (H) Portion	Default mask (Decimal)	Number of possible networks and hosts per network
A	0-127	00000000 - 01111111	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts ($2^{24}-2$)
B	128-191	10000000 - 10111111	N.N.H.H	255.255.0.0	16,384 Nets (2^{14}) 65,534 hosts ($2^{16}-2$)
C	192-223	11000000 - 11011111	N.N.N.H	255.255.255.0	2,091,50 Nets (2^{21}) 254 hosts (2^8-2)
D	224-239	11100000 - 11101111	NA (Multicast)	-	-
E	240-255	11110000 - 11111111	NA (Experimental)	-	-

- Class C addresses are assigned to small networks.
- The default mask for Class C addresses is 255.255.255.0 or /24 in CIDR notation.
- This default mask means that the first three octets represent the network portion, and the last octet represents the host portion.
- **Example** :- If an IP address is 192.168.0.1 with the default mask 255.255.255.0, the network portion is 192.168.0, and the host portion is 1.

Some examples of a class full addressing

Question :- A host in a class C network has been assigned an IP-address 192.168.17.9 , find the number of addresses , the first and the last addresses ?

Answer :- In a class C network, the first three octets are used for network identification, while the last octet represents the host address. Let's analyze the given information:

IP address: 192.168.17.9

To determine the number of addresses in the block, we need to consider the class C network and its default mask.

Class C default mask: 255.255.255.0

In class C, the host address portion has 8 bits (1 octet). This means there are $2^8 = 256$ possible combinations of host addresses within the network. (Total number of addresses not usable address)

However, in every network, two addresses are reserved:

- The first address represents the **network address**.
- The last address represents the **broadcast address**.

So, the total number of **usable** addresses in a class C network is $256 - 2 = 254$.

Now, let's find the first address and the last address :-

Given IP address: 192.168.17.9

The network address is obtained by setting the host portion to all zeros: **Network address: 192.168.17.0**

The last address is obtained by setting the host portion to all ones: **Last address (broadcast address): 192.168.17.255**

Therefore, in the given class C network, there are 254 addresses in the block (excluding the network and broadcast addresses), the first address is 192.168.17.0, and the last address is 192.168.17.255.

Question 2 :- An address in a block is given as 185.28.17.9 , find the number of addresses in the block , and find the first and the last addresses ?

Answer :- In a class B network, the first two octets are used for network identification, while the last two octet represents the host address. Let's analyze the given information

IP address: 185.28.17.9

Since the first octet of the IP address (185) falls within the range of 128 to 191, it indicates that the IP address belongs to Class B.

In Class B, the first two octets are used for network identification, and the remaining two octets represent the host address.

To determine the number of addresses in the block, we need to consider the class B network.

Class B default mask: 255.255.0.0

In Class B, the host address portion has 16 bits (2 octets). This means there are $2^{16} = 65,536$ possible combinations of host addresses within the network.

Similarly to Class C, two addresses are reserved:

- The first address represents the **network address**.
- The last address represents the **broadcast address**.

So, the total number of **usable** addresses in a class B network is $65,536 - 2 = 65,534$.

Now, let's find the first address and the last address:

Given IP address: 185.28.17.9

The network address is obtained by setting the host portion to all zeros: **Network address: 185.28.0.0**

The last address is obtained by setting the host portion to all ones: **Last address (broadcast address): 185.28.255.255**

Therefore, in the given class B network, there are 65,534 **usable** addresses in the block (excluding the network and broadcast addresses), the first address is 185.28.0.0, and the last address is 185.28.255.255.

Example – 3 :-

QUESTION

An organization follows class A for their internal network. One of the hosts in the network has an IP address 10.200.240.4. Find the number of addresses, the network address, and the broadcast address of the organization's network.

SOLUTION:

Class A Network

N.H.H (255.0.0.0 or /8)

10.200.240.4

This network: 10.0.0.0 - 10.255.255.255

Number of addresses: $2^{24} = 16,777,216$

Number of usable addresses: $16,777,216 - 2 = 16,777,214$

First Address: 10.0.0.0 (Network Address)

Last Address: 10.255.255.255 (Broadcast Address)

Network and Broadcast Address

Network Address: Imagine you live in a big neighborhood, and each house has a unique address. In the world of networking, the network address is like the address of your entire neighborhood. It identifies the **specific network** that your computer or device is connected to.

Just like every house in your neighborhood has the same neighborhood name, all devices within the same network share the same network address. It helps devices in the network recognize and communicate with each other.

Think of it this way: if you want to send a letter to someone in your neighborhood, you only need to write down the neighborhood name (network address) and their house number (host address) to ensure the letter reaches the correct house.

Broadcast Address: Now, imagine you want to send a message to everyone in your neighborhood at the same time. You can't individually deliver the message to each house; instead, you send a broadcast message that reaches every house in the neighborhood.

In networking, a broadcast address is like a special address that allows you to send a message to all devices within a network at once. It ensures that the message is received by every device within that network.

To do this, the broadcast address uses a special pattern in the host address part of an IP address. It sets all the host bits to their maximum value, which is usually represented as a sequence of 1s.

So, when a device receives a message with the broadcast address, it understands that the message is meant for everyone in the network and processes it accordingly.

Understanding these concepts helps devices in a network communicate with each other efficiently, whether it's sending messages to specific devices using their unique addresses or broadcasting messages to reach everyone within the network.

CLASS LESS IP ADDRESSING

Classless IP addressing, also known as Classless Inter-Domain Routing (CIDR), is an addressing scheme that provides a more flexible and efficient way of allocating IP addresses compared to classful IP addressing. In classless IP addressing, the division of IP addresses into predefined classes (Class A, B, or C) is abandoned, and the allocation is based on variable-length subnet masks (VLSM).

Here are some key points about classless IP addressing and how it solves the problems of classful IP addressing :-

1. **Flexibility in Address Allocation:** Classless IP addressing allows for more efficient utilization of IP address space. Instead of being constrained by fixed class boundaries, CIDR enables the allocation of IP addresses based on the specific needs of networks, allowing for custom-sized subnets.
2. **Variable-Length Subnet Masks (VLSM):** With CIDR, subnet masks can have varying lengths. A subnet mask determines the network and host portions of an IP address. By using longer or shorter subnet masks, network administrators can create subnets of different sizes to accommodate varying numbers of hosts within a network. This results in more efficient allocation of IP addresses.
3. **Efficient Address Space Allocation:** Classless IP addressing solves the problem of address space wastage that existed in classful addressing. Classful addressing had predefined address space for each class, resulting in inefficient allocation. CIDR enables more precise allocation of IP addresses, avoiding large blocks of unused or unallocated addresses.

For example, consider the following scenario:

- In classful addressing, a network requiring only 50 hosts would be allocated a Class C address block, which provides 254 usable addresses. This results in wastage of over 200 addresses.
- In classless addressing (CIDR), a network needing 50 hosts can be assigned a subnet with a mask of /26, which provides exactly 62 usable addresses. This allows for more efficient utilization of the address space.

Subnetting is the process of dividing a larger network into smaller subnetworks called subnets. It involves breaking down a network into more manageable and efficient segments to enhance network performance, security, and address allocation.

Imagine you have a big toy box filled with different types of toys. You want to keep your toys organized, so you decide to divide the box into smaller sections. Each section will have a different type of toy.

Subnetting is similar to dividing your toy box. Instead of toys, we are talking about computer networks. When computers are connected in a network, they have special numbers called IP addresses. These IP addresses help computers find and talk to each other.

Subnetting is a way to divide a big computer network into smaller parts, just like dividing your toy box. This helps keep things organized and makes it easier for computers to communicate with each other.

Imagine you have a group of friends, and you want to divide them into smaller teams. Each team will have a different team number. Subnetting works the same way. It divides the big network into smaller parts called subnets. Each subnet has its own subnet number.

By using subnetting, we can make sure that computers in one subnet can talk to each other easily, just like friends on the same team. It also helps computers send data more efficiently within their own subnet without overwhelming the whole network.

So, subnetting is like dividing a big computer network into smaller sections, similar to dividing toys or friends into groups. It helps computers stay organized and communicate better with each other.

Why Subnetting?

1. **Enhanced Network Performance:** Subnetting allows for better network performance by reducing network

congestion and improving overall data transmission efficiency. It enables local communication within smaller subnets, minimizing unnecessary traffic across the entire network.

2. **Improved Security:** Subnetting enhances network security by creating logical boundaries between different departments or user groups. It enables the implementation of security measures, such as firewalls or access control lists (ACLs), at the subnet level to control and monitor network traffic.
3. **Efficient Address Allocation:** Subnetting optimizes the utilization of IP addresses by assigning smaller blocks of addresses to each subnet. It prevents address wastage and allows for more efficient management of IP address space.

WHAT ARE SUBNETS ?

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So, subnetting is like dividing a big computer network into smaller sections, similar to dividing toys or friends into groups. It helps computers stay organized and communicate better with each other.

WHAT ARE SUBNET MASKS

A subnet mask is like set of rules that helps computers figure out which part of an address is for the network and which part is for the individual computer.

Think of the address as a big house with different rooms. The subnet mask tells us which rooms belong to the whole house (the network) and which rooms belong to each person (the individual computer).

For example, let's say we have the address 192.168.0.10 and the subnet mask 255.255.255.0. The subnet mask tells us that the first three numbers (192.168.0) are for the network, and the last number (10) is for the individual computer.

It's like having a map that shows which parts of the address are important for finding the right network and computer. The subnet mask helps computers understand where to send information and how to talk to each other within the network.

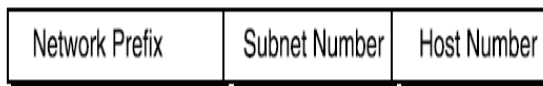
So, a subnet mask is like a special mask or map that helps computers know which part of an address is for the network and which part is for the individual computer. It's like having a secret code to open the right doors and find the right place to send information.

EXTENDED NETWORK PREFIX

Two-Level Classful Hierarchy



Three-Level Subnet Hierarchy



Subnetting is a technique used in computer networking to divide a single IP network into multiple smaller subnetworks, or subnets. It allows for more efficient utilization of IP addresses and provides better network management and control.

In the traditional classful IP addressing scheme, the IP address space was divided into three classes: Class A, Class B, and Class C. Each class had a fixed number of network bits and host bits. This two-level hierarchy provided a limited number of network addresses and a fixed number of hosts per network.

To overcome the limitations of the classful addressing scheme and address the mentioned problems, subnetting was introduced. Subnetting adds another level of hierarchy by further dividing the host portion of the IP address into two parts: the subnet number and the host number on that subnet.

By allocating a portion of the host bits for the subnet number, organizations can create smaller subnets within a larger network. This allows for more flexibility in network design, efficient allocation of IP addresses, and improved network performance.

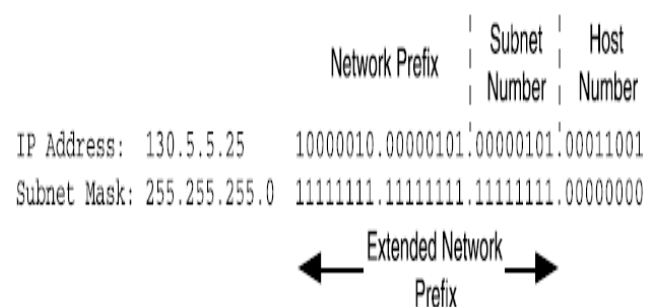
With subnetting, organizations can create subnets based on their specific requirements. They can allocate different subnet sizes depending on the number of hosts required in each subnet. This results in better utilization of IP address space, as smaller subnets can be assigned where fewer hosts are needed, while larger subnets can be used where more hosts are expected.

In simpler terms, the subnet mask is used to determine which part of an IP address is the network address and which part is the host address. **The subnet mask is a series of numbers that has a one-to-one correspondence with the IP address. Each bit in the subnet mask is either set to 1 or 0.**

When a bit in the subnet mask is set to 1, it means that the corresponding bit in the IP address is part of the network address. When a bit is set to 0, it means that the corresponding bit in the IP address is part of the host address.

For example, let's say we have the IP address 130.5.0.0 with a subnet mask of 255.255.255.0. In this case, the first three octets (130.5.0) represent the network address, and the last octet (0) represents the host address. The subnet mask helps us determine this division by using 1s for the network portion and 0s for the host portion.

This division is important because it allows computers and routers to understand which devices are on the same network and can directly communicate with each other. The subnet mask helps define the boundaries of the network and host parts of an IP address, making it easier for devices to route data packets and manage network traffic.



THE PREFIX LENGTH NOTATION

In computer networks, there are special codes called subnet masks and prefix lengths that help us understand and find the right network for sending information. These codes are used in modern routing protocols, which are like maps for computers to navigate and communicate with each other.

Now, let's talk about the extended network prefix length and how it's used in routing protocols.

The extended network prefix length is a way of representing the subnet mask, which tells us which part of an IP address belongs to the network. Instead of using the traditional subnet mask format, modern routing protocols often use a shorter and simpler notation called the prefix length.

The prefix length is the number of continuous "one" bits in the subnet mask. It shows us how many bits are important for identifying the network.

For example, let's say we have the network address 130.5.5.25 with a subnet mask of 255.255.255.0. This means the first 24 bits (the first three numbers) are part of the network, and the last 8 bits (the last number) are for individual devices.

To make things easier, we can express the same information using the prefix length notation. In this case, we write the network address followed by a slash and the prefix length, which is 24. So, 130.5.5.25 with a subnet mask of 255.255.255.0 can be written as 130.5.5.25/24.

This notation is more compact and easier to understand compared to writing out the long subnet mask with all the numbers. It helps us quickly identify the network part of an IP address and simplifies communication between devices.

130.5.5.25	10000010.00000101.00000101.00011001
255.255.255.0	11111111.11111111.11111111.00000000

or

130.5.5.25/24	10000010.00000101.00000101.00011001
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BORROWING BITS FROM THE HOST

When we talk about borrowing bits from the host, it means taking some bits that are originally used for identifying individual hosts and repurposing them to create additional subnets.

In computer networks, IP addresses are made up of two parts: the network portion and the host portion. The network portion identifies the network itself, while the host portion identifies individual devices within that network.

When we need to create more subnets within a network, we may need to borrow bits from the host portion and use them to create additional subnets. This allows us to divide the network into smaller, more manageable parts.

By borrowing bits from the host portion, we reduce the number of available host addresses because some of those bits are now used for subnetting. This trade-off allows for more flexibility in network design and more efficient use of IP address space.

For example, let's say we have an IP address range of 192.168.0.0/24, which means there are 24 bits reserved for the network and 8 bits for hosts. If we want to create more subnets, we can borrow some bits from the host portion, let's say 3 bits.

By borrowing 3 bits, we increase the number of subnets we can create, but we decrease the number of available host addresses within each subnet. The number of subnets is calculated using the formula 2^N , where N is the number of borrowed bits.

So, by borrowing 3 bits, we can create $2^3 = 8$ subnets.

However, keep in mind that borrowing bits also reduces the number of available host addresses per subnet. The remaining bits in the host portion will determine the number of hosts per subnet. The formula $2^M - 2$ is used to calculate the number of usable host addresses, where M is the number of remaining host bits.

Borrowing bits allows for subnetting and creating smaller subnetworks within a larger network, providing more flexibility and efficient allocation of resources.

NUMBER OF SUBNETS AND NUMBER OF HOSTS PER SUBNET

The calculation of subnets and the number of hosts per subnet in an IP network involves a solid understanding of the concepts of subnetting and IP addressing. Here is a basic guide:

1. **Number of Subnets:** The number of subnets is determined by the number of bits borrowed from the host portion of the network for subnetting. If you borrow n bits, you can have 2^n subnets. For example, in a Class C network (default subnet mask of 255.255.255.0), if you borrow 2 bits, you can create 2^2 or 4 subnets.
2. **Number of Hosts Per Subnet:** The number of valid hosts per subnet is $2^n - 2$ where n is the number of bits remaining for host addressing after subnetting. The "-2" accounts for the network address and broadcast address which are not usable for hosts. For example, if you have 6 bits remaining for host addressing (after borrowing 2 bits for subnetting in a Class C network), you can have $2^6 - 2$ or 62 hosts per subnet.

Remember that IP version 4 (IPv4) addresses are 32-bit addresses. A typical IPv4 address looks like this: 192.168.1.1. The subnet mask (e.g., 255.255.255.0) is used to divide the IP address into the network and host addresses.

Here's an example with a Class C IP address:

Let's say we have a Class C network with a base IP address of 192.168.1.0 and a subnet mask of 255.255.255.0 (or /24 in CIDR notation, meaning the first 24 bits are used for network addressing). If we want to create additional subnets, we can borrow bits from the host part of the address.

If we borrow 2 bits, our subnet mask becomes 255.255.255.192 (or /26 in CIDR notation).

1. To calculate the **number of subnets**, we use the formula 2^n , where n is the number of borrowed bits. So, 2^2 equals 4 subnets.
2. To calculate the **number of hosts per subnet**, we use the formula $2^n - 2$, where n is the number of bits left for the host address. Since we borrowed 2 bits, we have 6 bits left for hosts (as there are 8 bits in total for the last octet of the address). So, $2^6 - 2$ equals 62 hosts.

Please note, however, that the method of calculation can differ based on whether you're using fixed-length or variable-length subnet masking, and whether you're using private or public IP addresses. The "-2" in the host calculation is also a traditional approach that excludes the network and broadcast addresses, but in some modern systems, all addresses can potentially be used.

EXAMPLE :- Suppose our IP address is 192.168.2.64/26, figure out the usable address, the start or network address and the end or broadcast address, find the number of subnets, the number of the host per subnet?

The IP address 192.168.2.64/26 means that the network portion is the first 26 bits of the address, and the remaining 6 bits are for host addresses within the subnet.

To calculate the details for this address:

1. **Subnet Mask:** For a /26 CIDR, the subnet mask is 255.255.255.192. The /26 means that the first 26 bits are used for the network address and the remaining 6 bits ($32 - 26$) are used for the host addresses.
2. **Network Address (Start Address):** This is the first IP address in the subnet and is used to identify the subnet itself. It's typically the IP address where the host bits are all set to 0. For 192.168.2.64/26, the network address is 192.168.2.64.
3. **Broadcast Address (End Address):** This is the last IP address in the subnet and is used for sending data to all hosts on the

subnet simultaneously. It's the IP address where the host bits are all set to 1. For 192.168.2.64/26, the broadcast address is 192.168.2.127.

4. **Usable Addresses:** These are the IP addresses that can be assigned to the hosts within the subnet, which are all the addresses between the network address and the broadcast address. For 192.168.2.64/26, the usable addresses range from 192.168.2.65 to 192.168.2.126.

5. **Number of Subnets:** Since 192 is a class network, we first have a /24 network and later changed into /26 subnets, you are effectively borrowing 2 bits (26 - 24) for subnetting. Therefore, using the formula 2^n , where n is the number of borrowed bits, you get $2^2 = 4$ subnets. Note that this calculation is based on the assumption that you started with a /24 network, which is typically the case for a Class C IP address like this one.

6. **Number of Hosts per Subnet:** This is calculated by the formula $2^n - 2$, where n is the number of bits left for the host address. Since we have 6 bits for hosts (32 total - 26 for network = 6 for hosts), you get $2^6 - 2 = 62$ hosts per subnet.

CLASS SUB-NETTING EXAMPLE :- subnet the IP address 216.21.5.0 in to 30 hosts in each subnet ?

The IP address given is 216.21.5.0, and we want each subnet to support around 30 hosts.

Here are the steps:

Find the appropriate subnet mask:

First, we need to determine how many bits are needed to support around 30 hosts per subnet. The formula to calculate the number of hosts is $2^n - 2$ (where n is the number of host bits), but we'll simplify it to 2^n for an easier approximation.

Looking at powers of 2, we find that 2^5 is 32. So, we need 5 bits for hosts because that will give us 32 addresses (close to 30). That leaves us with 3 bits for subnetting (8 total bits in an octet - 5 bits for hosts = 3 bits for subnetting).

Therefore, the subnet mask should be 255.255.255.224 (in binary, this last octet is 11100000, which is 3 bits for subnetting and 5 bits for hosts).

Calculate the number of subnets:

We borrowed 3 bits for subnetting, so we can calculate the number of subnets with the formula 2^n (where n is the number of subnet bits). So, 2^3 equals 8. Therefore, we have 8 subnets.

Calculate the network address (start address) and the broadcast address (end address) for each subnet:

The subnets would be broken down like this (only showing the last octet for simplicity):

- Subnet 1: Network 0, Usable 1-30, Broadcast 31
- Subnet 2: Network 32, Usable 33-62, Broadcast 63
- Subnet 3: Network 64, Usable 65-94, Broadcast 95
- Subnet 4: Network 96, Usable 97-126, Broadcast 127
- Subnet 5: Network 128, Usable 129-158, Broadcast 159
- Subnet 6: Network 160, Usable 161-190, Broadcast 191
- Subnet 7: Network 192, Usable 193-222, Broadcast 223
- Subnet 8: Network 224, Usable 225-254, Broadcast 255

So for example, in the first subnet, you could assign host IP addresses from 216.21.5.1 to 216.21.5.30. The network address would be 216.21.5.0, and the broadcast address would be 216.21.5.31.

And in the second subnet, you could assign host IP addresses from 216.21.5.33 to 216.21.5.62, and so on for the remaining subnets.

Remember that the network and broadcast addresses are traditionally not assigned to hosts. However, in some modern systems, all addresses can potentially be used.

Also note that the exact ranges may depend on the specifics of your network architecture and the equipment you're using. Always refer to the documentation and specifications of your specific networking equipment when configuring subnets.

CLASS C SUB-NETTING EXAMPLE :- subnet the IP address 196.10.20.0 in to 52 hosts in each subnet ?

let's subnet the IP address 196.10.20.0 so that there are approximately 52 hosts in each subnet. We'll do this in a simple way:

Find the appropriate subnet mask:

First, we need to determine how many bits are needed to support approximately 52 hosts per subnet. The formula to calculate the number of hosts is $2^n - 2$, where n is the number of host bits, but to keep it simple, we'll use 2^n for a rough approximation.

Looking at powers of 2, we find that 2^6 is 64, which is the next number that can accommodate at least 52 hosts. So, we need 6 bits for hosts, because that will give us 64 addresses, which is more than enough for our needs. That leaves us with 2 bits for subnetting (8 total bits in an octet - 6 bits for hosts = 2 bits for subnetting).

Therefore, the subnet mask should be 255.255.255.192 (in binary, this last octet is 11000000, which is 2 bits for subnetting and 6 bits for hosts).

Calculate the number of subnets:

We borrowed 2 bits for subnetting, so we can calculate the number of subnets with the formula 2^n (where n is the number of subnet bits). So, 2^2 equals 4. Therefore, we have 4 subnets.

Calculate the network address (start address) and the broadcast address (end address) for each subnet:

The subnets would be broken down like this (only showing the last octet for simplicity):

- Subnet 1: Network 0, Usable 1-62, Broadcast 63
- Subnet 2: Network 64, Usable 65-126, Broadcast 127
- Subnet 3: Network 128, Usable 129-190, Broadcast 191
- Subnet 4: Network 192, Usable 193-254, Broadcast 255

So for example, in the first subnet, you could assign host IP addresses from 196.10.20.1 to 196.10.20.62. The network address would be 196.10.20.0, and the broadcast address would be 196.10.20.63.

And in the second subnet, you could assign host IP addresses from 196.10.20.65 to 196.10.20.126, and so on for the remaining subnets.

Remember that the network and broadcast addresses are traditionally not assigned to hosts. However, in some modern systems, all addresses can potentially be used.

Also note that the exact ranges may depend on the specifics of your network architecture and the equipment you're using. Always refer to the documentation and specifications of your specific networking equipment when configuring subnets.

CLASS B SUB-NETTING EXAMPLE :-

let's subnet the IP address 150.15.0.0 to accommodate approximately 500 hosts in each subnet, and we'll do this in a simple way:

Find the appropriate subnet mask:

We first need to figure out how many bits are needed to support approximately 500 hosts per subnet. We could use the formula $2^n - 2$ (where n is the number of host bits), but let's keep it simple and use 2^n for an easy approximation.

Looking at powers of 2, we find that 2^9 is 512. This is the smallest power of 2 that's greater than 500. So, we need 9 bits for hosts because that

will give us 512 addresses which is more than our requirement of 500. That leaves us with 7 bits for subnetting (16 total bits in the last two octets - 9 bits for hosts = 7 bits for subnetting).

Therefore, the subnet mask should be 255.255.254.0 (in binary, the last two octets are 11111110 00000000, which is 7 bits for subnetting and 9 bits for hosts).

Calculate the number of subnets:

We borrowed 7 bits for subnetting, so we can calculate the number of subnets with the formula 2^n (where n is the number of subnet bits). So, 2^7 equals 128. Therefore, we have 128 subnets.

Calculate the network address (start address) and the broadcast address (end address) for each subnet:

The subnets would be broken down like this (only showing the last two octets for simplicity):

- Subnet 1: Network 0.0, Usable 0.1-1.254, Broadcast 1.255
- Subnet 2: Network 2.0, Usable 2.1-3.254, Broadcast 3.255
- Subnet 3: Network 4.0, Usable 4.1-5.254, Broadcast 5.255
- And so on until Subnet 128.

So for example, in the first subnet, you could assign host IP addresses from 150.15.0.1 to 150.15.1.254. The network address would be 150.15.0.0, and the broadcast address would be 150.15.1.255.

And in the second subnet, you could assign host IP addresses from 150.15.2.1 to 150.15.3.254, and so on for the remaining subnets.

Remember that the network and broadcast addresses are traditionally not assigned to hosts. However, in some modern systems, all addresses can potentially be used.

Also note that the exact ranges may depend on the specifics of your network architecture and the equipment you're using. Always refer to the documentation and specifications of your specific networking equipment when configuring subnets.

CLASS A SUB-NETTING EXAMPLE

Let's use a Class A network of 10.0.0.0 and subnet it to accommodate around 2000 hosts in each subnet.

Default Subnet Mask:

For a Class A network, the default subnet mask is 255.0.0.0. This means that by default, the first 8 bits (1 octet) are used for the network portion of the address, and the remaining 24 bits (3 octets) are available for host addresses.

Find the appropriate subnet mask:

If we want to support approximately 2000 hosts per subnet, we need to determine how many host bits are required. Looking at powers of 2, we find that 2^{11} is 2048, which is the smallest power of 2 that's greater than 2000. So, we need 11 bits for hosts because that will give us 2048 addresses, which is more than our requirement of 2000. That leaves us with 13 bits for subnetting (24 total bits in the last three octets - 11 bits for hosts = 13 bits for subnetting).

Therefore, the subnet mask should be 255.248.0.0 (in binary, the last three octets are 11111000 00000000 00000000, which is 13 bits for subnetting and 11 bits for hosts).

Calculate the number of subnets:

We borrowed 13 bits for subnetting, so we can calculate the number of subnets with the formula 2^n (where n is the number of subnet bits). So, 2^{13} equals 8192. Therefore, we have 8192 subnets.

Calculate the network address (start address) and the broadcast address (end address) for each subnet:

Each subnet has 2^{11} or 2048 addresses. Out of these, one is used as the network address and another one is used as the broadcast address. The rest (2046 addresses) are available for hosts.

The network addresses of the first few subnets would look something like this:

- Subnet 1: Network 10.0.0.0, Usable 10.0.0.1 to 10.0.7.254, Broadcast 10.0.7.255
- Subnet 2: Network 10.0.8.0, Usable 10.0.8.1 to 10.0.15.254, Broadcast 10.0.15.255
- Subnet 3: Network 10.0.16.0, Usable 10.0.16.1 to 10.0.23.254, Broadcast 10.0.23.255
- And so on until Subnet 8192.

Remember that the network and broadcast addresses are traditionally not assigned to hosts. However, in some modern systems, all addresses can potentially be used.

Also note that the exact ranges may depend on the specifics of your network architecture and the equipment you're using. Always refer to the documentation and specifications of your specific networking equipment when configuring subnets.



